

PRIMARY 5 BASIC SCIENCE & TECHNOLOGY SCHEME OF WORK

FIRST TERM

Week	Topic	Content
1	Living Things and Their Environment	Habitats and adaptations; Relationships between organisms; Food chains and food webs; Interdependence in nature
2	Plant Structure and Functions	Detailed study of roots, stems, leaves, flowers and fruits; Photosynthesis process; Transport in plants; Reproduction in flowering plants
3	Animal Classification	Vertebrates and invertebrates; Major animal groups; Characteristics of each group; Local examples of animals in each class
4	Human Body: The Circulatory System	The heart and blood vessels; Functions of blood; The circulation process; Keeping the heart healthy; Blood groups
5	MIDTERM EXAMINATION	Assessment of topics covered in weeks 1-4
6	MIDTERM BREAK	Rest and relaxation period
7	Human Body: The Digestive System	The alimentary canal; Digestion process; Functions of digestive organs; Absorption of nutrients; Digestive health
8	Human Body: The Skeletal System	Major bones in the body; Functions of the skeleton; Joints and movement; Bone health and care; Common bone problems
9	Matter and Materials	States of matter - solid, liquid, gas; Properties of materials; Changing states; Materials around us
10	Mixtures and Separation	Types of mixtures; Methods of separation - filtration, evaporation, sedimentation; Practical applications; Pure substances
11	REVISION	Review of all topics covered; Practice exercises; Question and answer sessions
12	EXAMINATION	End of term assessment

13	CLOSING	Report card distribution; Vacation assignment
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PRIMARY 5 BASIC SCIENCE & TECHNOLOGY LESSON NOTES

FIRST TERM

WEEK 1: LIVING THINGS AND THEIR ENVIRONMENT

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define the term "Habitat" and list three types of habitats.
2. Explain the meaning of adaptation with one example.
3. Draw a simple food chain showing a producer and consumers.
4. Describe how living things depend on each other (Interdependence).

CONTENT

1. Habitats and Adaptations

- **Habitat:** A habitat is the natural home or environment where a plant or animal lives and grows. Think of it like your own house. Your house gives you a place to sleep, food from the kitchen, and water to drink. In the same way, a habitat provides an animal or plant with food, water, air, shelter, and a place to raise its young.
 - **Aquatic Habitat:** This is a water home. It can be fresh water like our rivers (e.g., River Niger), ponds, and lakes, or salt water like the ocean (e.g., Atlantic Ocean by Lagos). Fish, crabs, water lilies, and hippopotamuses live here.
 - **Terrestrial Habitat:** This is a land home. It has different types:
 - **Grasslands/Savannah:** Open fields with lots of grass, like the places where we see cattle grazing. Animals here include lions, antelopes, and grasshoppers.
 - **Forests:** Places with many trees, like the Okomu Forest. Animals here include monkeys, snakes, and many birds.
 - **Deserts:** Very dry and hot places with little water, like some parts of Northern Nigeria. Animals here include camels and scorpions.

- **Arboreal Habitat:** This is a special home in the trees. Some animals spend most of their time there. Examples are birds (they build nests in trees), squirrels, and some monkeys.



- **Adaptation:** This is a special feature or skill a living thing has that makes it perfectly suited to live in its particular habitat. It helps it find food, water, and shelter, and protect itself.
 - **Example 1:** A fish has gills to breathe underwater and fins and a streamlined body to swim easily. Its body is adapted for a water habitat.
 - **Example 2:** A camel has wide feet to walk on sand without sinking, long eyelashes to keep sand out of its eyes, and a hump that stores fat for energy when there is no food or water in the desert. Its body is adapted for a desert habitat.
 - **Example 3:** A bird has wings to fly, light bones, and a beak shaped for eating its specific food (e.g., a hawk has a hooked beak for tearing meat).

2. Relationships Between Organisms Living things in a habitat do not live alone; they interact with each other.

- **Predation:** This is a "hunter and hunted" relationship. One animal (the predator) hunts, kills, and eats another animal (the prey). Example: A lizard (predator) eats a fly (prey).

- **Competition:** This happens when two or more living things need the same thing, but there is not enough for everyone. They "compete" or struggle for it. Example: Two mango trees in a yard compete for sunlight, water, and nutrients from the soil.

3. Food Chains and Food Webs

- **Food Chain:** This is a simple, straight line that shows how energy in food passes from one living thing to another. It always starts with energy from the sun.
 - **Producer:** A green plant. It produces its own food using sunlight (photosynthesis). It is the first and most important part of any food chain. Example: Grass, maize plant.
 - **Consumer:** An animal that eats other living things.
 - **Primary Consumer:** Eats the producer. It is usually a plant-eating animal (herbivore). Example: A grasshopper eats grass.
 - **Secondary Consumer:** Eats the primary consumer. It is usually a meat-eating animal (carnivore). Example: A frog eats the grasshopper.
 - **Tertiary Consumer:** Eats the secondary consumer. Example: A snake eats the frog.

Simple Food Chain: Sun → Grass (Producer) → Grasshopper (Primary Consumer) → Frog (Secondary Consumer) → Snake (Tertiary Consumer).



- **Food Web:** In a real habitat, feeding relationships are not single straight lines. Most animals eat more than one type of food. A Food Web is many food chains linked together like a spider's web. It shows that all living things in a habitat are connected.

4. Interdependence in Nature This means that all living things in a habitat need each other to survive. They depend on one another.

- **Animals depend on plants:** For food (fruits, leaves) and for oxygen to breathe.
- **Plants depend on animals:** For carbon dioxide (which animals breathe out) to make their own food. They also depend on animals for pollination (bees) and for manure (animal waste) which makes the soil fertile.
- If one part of this connection is removed (e.g., if all bees disappear), it affects many other plants and animals. It's like a puzzle; every piece is important.

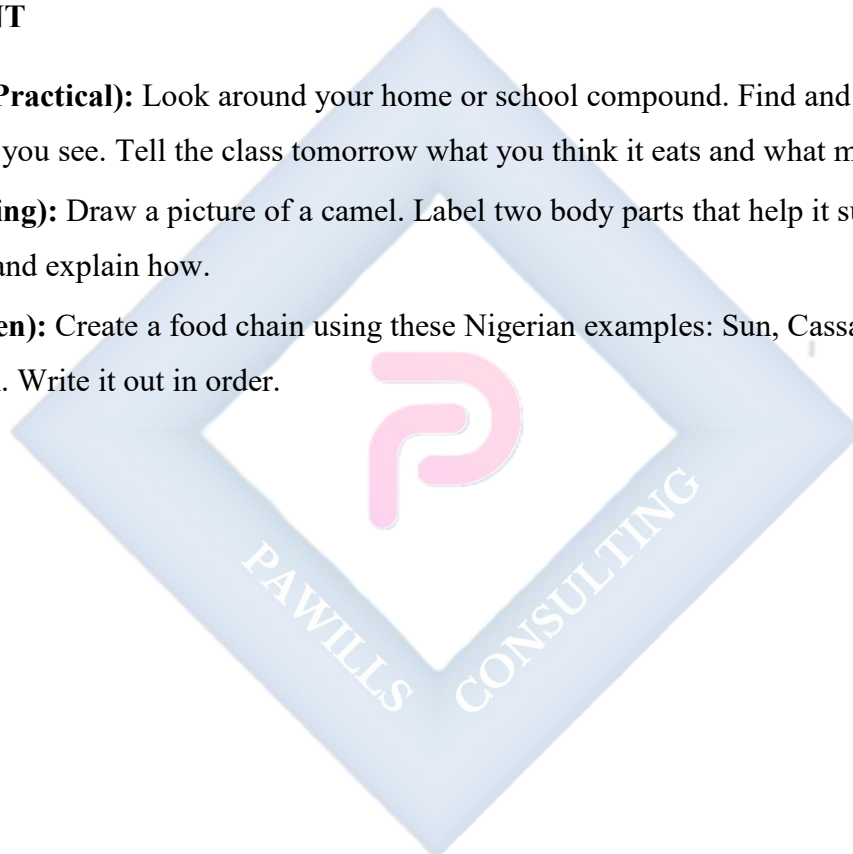
EVALUATION

1. What is a habitat? Give one example of an animal and its habitat.
2. Look at this animal: It has thick fur, a layer of fat under its skin, and large paws. What kind of habitat (hot or cold) do you think it is adapted to live in? Why?

3. A fish uses its _____ to swim and its _____ to breathe underwater. (Fill in the blanks).
4. Why must every food chain begin with a green plant (producer)?
5. Use these living things to draw a simple food chain: Sun, Maize Plant, Chicken, Hawk. Label the producer and consumers.
6. How do plants and animals help each other? Mention one way plants help animals and one way animals help plants.

ASSIGNMENT

1. **(Oral/Practical):** Look around your home or school compound. Find and name one animal you see. Tell the class tomorrow what you think it eats and what might eat it.
2. **(Drawing):** Draw a picture of a camel. Label two body parts that help it survive in the desert and explain how.
3. **(Written):** Create a food chain using these Nigerian examples: Sun, Cassava plant, Goat, Human. Write it out in order.



WEEK 2: PLANT STRUCTURE AND FUNCTIONS

OBJECTIVES By the end of the lesson, pupils should be able to:

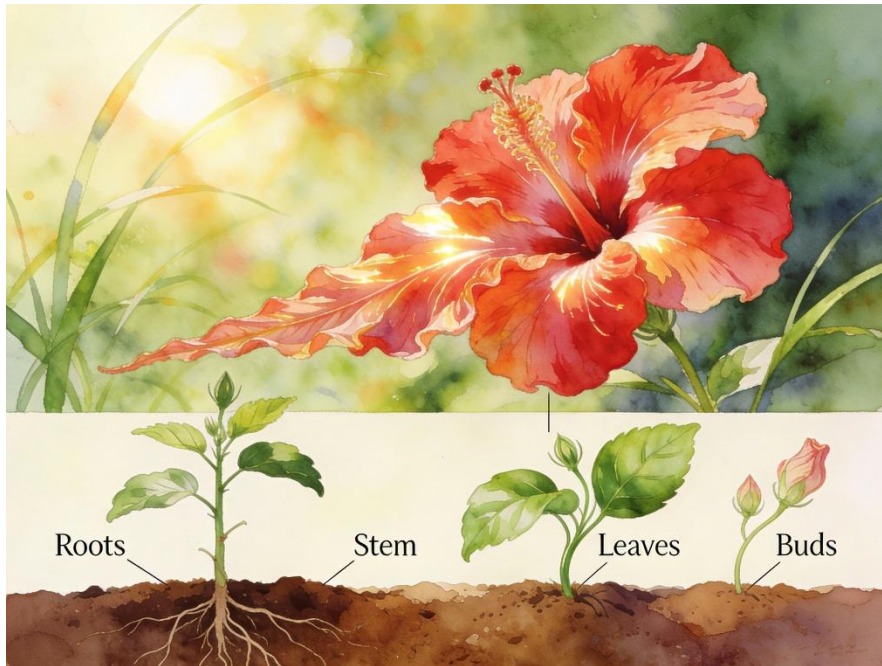
1. Identify the main parts of a plant (roots, stem, leaves, flowers).
2. Explain the process of photosynthesis simply.
3. Describe how water and food are transported in plants.
4. State the function of flowers in reproduction.

CONTENT

1. Parts of a Plant and Their Jobs A plant is like a factory with different departments, each with a special job.

- **Roots:** They are usually under the soil.
 - **Job 1:** They act like an anchor, holding the plant firmly in the ground so it doesn't fall over when the wind blows.
 - **Job 2:** They have tiny hair-like parts that absorb (suck up) water and dissolved minerals (plant food) from the soil.
- **Stem:** This is the plant's main support and transport highway.
 - **Job 1:** It holds up the leaves, flowers, and fruits towards the sunlight.
 - **Job 2:** It has tiny tubes inside that act like pipes. They carry water and minerals upwards from the roots to the leaves. They also carry food downwards from the leaves to other parts of the plant.
- **Leaves:** This is the plant's kitchen or food factory.
 - **Job:** They make food for the entire plant through a process called photosynthesis. Most leaves are broad, flat, and green to catch as much sunlight as possible.
- **Flower:** This is the plant's reproduction unit.
 - **Job:** Its main job is to produce seeds so that new plants can grow. Flowers are often bright and sweet-smelling to attract insects like bees and butterflies which help in pollination.
- **Fruit:** This is the seed protector.

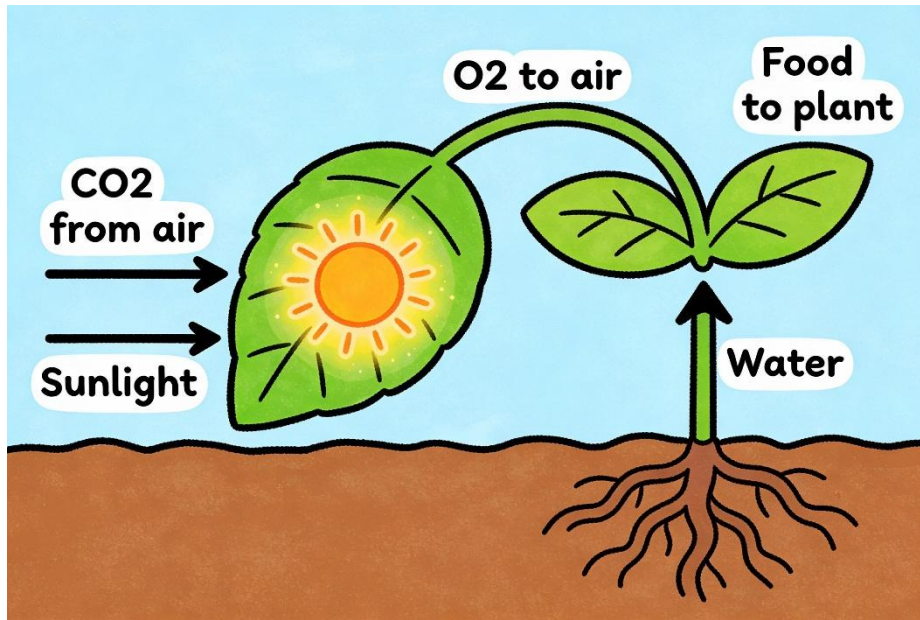
- **Job:** It grows from the flower after pollination and protects the seeds inside (like a mango protects its seed). It also helps in seed dispersal (spreading seeds).



2. Photosynthesis: How Plants Make Food "Photo" means light. "Synthesis" means putting together. Photosynthesis is the process where green plants use light energy to put together water and carbon dioxide to make their food (sugar).

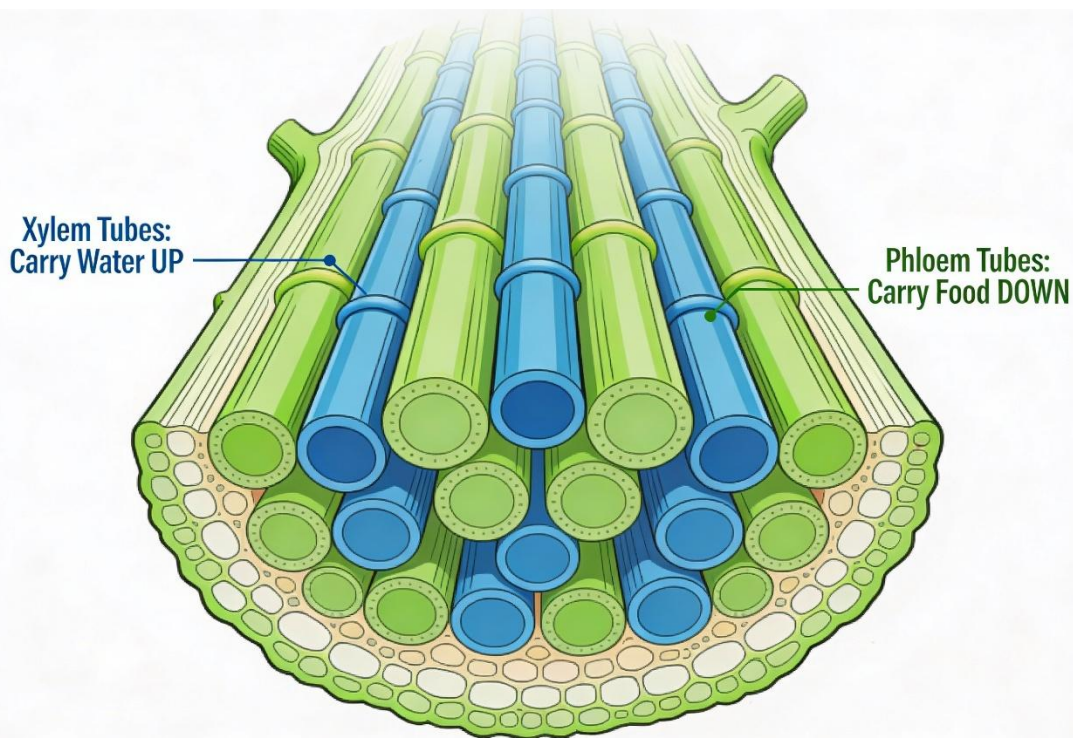
- **What the plant takes IN:**
 - **Water:** From the soil, through the roots.
 - **Carbon Dioxide (CO₂):** From the air, through tiny holes on the underside of leaves called stomata.
 - **Sunlight:** Energy captured by the green pigment chlorophyll in the leaves.
- **What the plant makes and gives OUT:**
 - **Sugar (Glucose):** This is the plant's food. It is used for energy and growth. Extra sugar is stored as starch (like in yam, cassava).
 - **Oxygen (O₂):** A very important gas that is released into the air. Animals and humans need oxygen to breathe!

Simple Word Equation for Photosynthesis: Sunlight Water + Carbon Dioxide → Sugar (Food) + Oxygen (Chlorophyll helps this happen)



3. Transport in Plants: The Plant's Pipeline System Inside the stem are two types of tiny tubes, like our veins and arteries.

- **Xylem (pronounced ZY-lem):** These are the water pipes. They carry water and minerals from the roots upwards to all parts of the plant.
- **Phloem (pronounced FLO-em):** These are the food pipes. They carry the sugary food made in the leaves downwards to the roots and other parts that need it (like growing fruits).



4.

Reproduction in Flowering Plants

- **Step 1 – Pollination:** Pollen grains (like tiny yellow dust) need to move from the male part of a flower (anther) to the female part (stigma) of the same or another flower. Insects (bees, butterflies) or wind help do this. When a bee visits a flower for nectar, pollen sticks to its body and is carried to the next flower.
- **Step 2 – Fertilization:** The pollen grain grows a tube down to the ovary and joins with an egg cell. This union is called fertilization.
- **Step 3 – Fruit and Seed Formation:** After fertilization, the ovary swells and becomes a fruit (like a mango, orange, or tomato). The fertilized eggs inside become seeds. The seeds can grow into new plants.

EVALUATION

1. Which part of the plant is like a kitchen? What does it produce there?
2. What are the two main jobs of the roots?
3. For photosynthesis, plants need three main things. Name them.
4. What two important things are produced during photosynthesis?

5. Why are flowers often brightly colored and sweet-smelling?
6. If you see a bee moving from flower to flower, what important process is it helping with?

ASSIGNMENT

1. **(Drawing):** Draw and color a flowering plant. Label these four parts: Root, Stem, Leaf, Flower.
2. **(Written):** In your own words, write one sentence to explain what photosynthesis is.
3. **(Practical):** With the help of an adult, gently pluck a leaf from a green plant (do not destroy the plant). Use clear tape to stick it in your notebook. Write one sentence about what the leaf does for the plant.



WEEK 3: ANIMAL CLASSIFICATION

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Differentiate between vertebrates and invertebrates.
2. List the five major groups of vertebrates.
3. Mention characteristics of mammals and birds.
4. Identify local examples of animals in each group.

CONTENT

1. Vertebrates and Invertebrates Scientists group animals based on whether they have a backbone or not.

- **Vertebrates:** These are animals that have a backbone (also called a spine or vertebral column) inside their bodies. The backbone is made of many small bones called vertebrae. It supports the body and protects the spinal cord (the main nerve cable). Examples: Human, goat, fish, snake, frog, chicken.
 - **How to remember:** "Vertebrate" sounds like "vertebra" which means backbone.
- **Invertebrates:** These are animals that do NOT have a backbone. Their bodies may be soft or have a hard outer shell. This group contains most of the animals on Earth! Examples: Fly, mosquito, earthworm, snail, spider, prawn, crab, scorpion.



2. Major Vertebrate Groups (5 Classes) Vertebrates are further divided into 5 main classes based on their body covering, how they breathe, and how they reproduce.

- **Mammals**

- **Body Covering:** Hair or fur on their skin.
- **Breathing:** With lungs.
- **Reproduction:** Give birth to live young ones (not eggs). The mother feeds her babies with milk from her body.
- **Other:** Warm-blooded (body temperature stays constant).
- **Local Examples:** Human being, cow, goat, dog, cat, bat, whale.

- **Birds**

- **Body Covering:** Feathers.
- **Breathing:** With lungs.
- **Reproduction:** Lay hard-shelled eggs. They build nests.
- **Other:** Have wings and beaks. Most can fly. Warm-blooded.
- **Local Examples:** Chicken, pigeon, eagle, duck, turkey.

- **Reptiles**

- **Body Covering:** Dry, scaly skin.
- **Breathing:** With lungs.
- **Reproduction:** Lay soft, leathery eggs on land.
- **Other:** Cold-blooded (body temperature changes with surroundings). Most crawl on their belly or on short legs.
- **Local Examples:** Lizard, snake, crocodile, turtle, chameleon.

- **Amphibians**

- **Body Covering:** Moist, smooth skin (no scales).
- **Breathing:** Young ones (tadpoles) breathe with gills in water. Adults breathe with lungs and through their skin on land.
- **Reproduction:** Lay soft eggs in water.
- **Other:** Cold-blooded. They live a "double life" – start in water, live on land as adults.

- **Local Examples:** Frog, toad.
- **Fish**
 - **Body Covering:** Wet scales.
 - **Breathing:** With gills (which take oxygen from water).
 - **Reproduction:** Lay many soft eggs in water.
 - **Other:** Have fins for swimming. Cold-blooded. Live only in water.
 - **Local Examples:** Tilapia, catfish, mackerel, sardine.



3. Looking at Invertebrates Invertebrates are many! Here are some common groups:

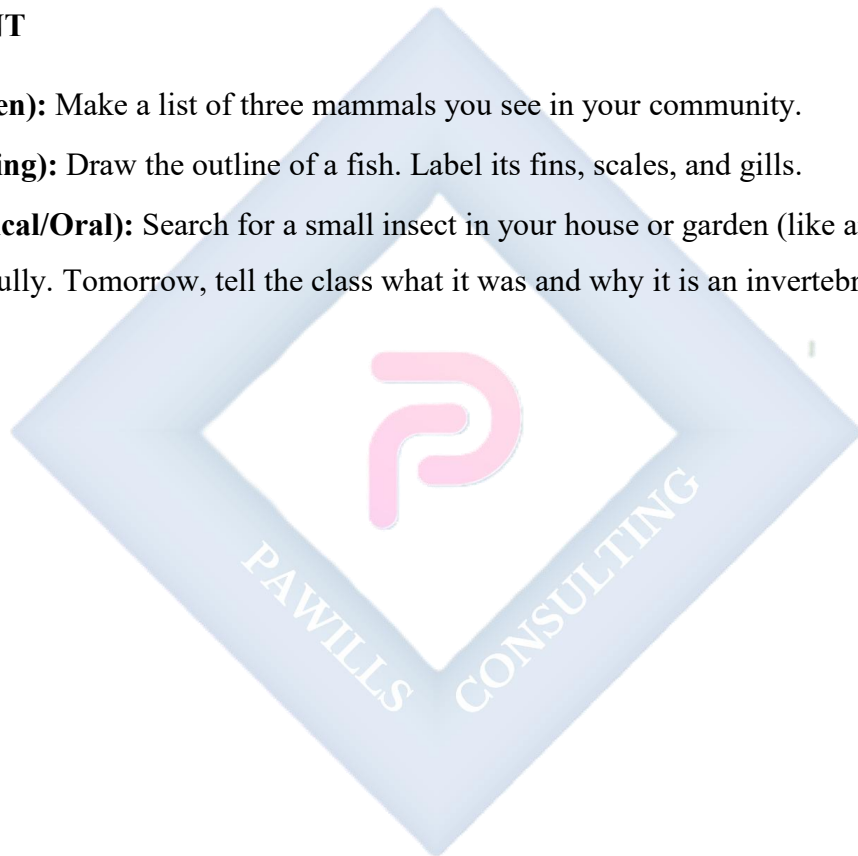
- **Insects:** Have 6 legs, body divided into head, thorax, abdomen. Example: Ant, butterfly, beetle.
- **Arachnids:** Have 8 legs. Example: Spider, scorpion.
- **Crustaceans:** Have hard shells, live in water. Example: Crab, prawn.
- **Mollusks:** Soft bodies, some have shells. Example: Snail, slug, octopus.
- **Worms:** Long, soft bodies. Example: Earthworm.

EVALUATION

1. What is the one big difference between a monkey and a housefly? (Think about backbone).
2. Name the five groups of vertebrates.
3. An animal has feathers and lays eggs. Which vertebrate group does it belong to?
4. Which group do humans belong to? Name two features of that group.
5. Why can a fish not live on land like a lizard? (Think about how it breathes).
6. Is a snail a vertebrate or an invertebrate? How do you know?

ASSIGNMENT

1. **(Written):** Make a list of three mammals you see in your community.
2. **(Drawing):** Draw the outline of a fish. Label its fins, scales, and gills.
3. **(Practical/Oral):** Search for a small insect in your house or garden (like an ant). Observe it carefully. Tomorrow, tell the class what it was and why it is an invertebrate.



WEEK 4: HUMAN BODY: THE CIRCULATORY SYSTEM

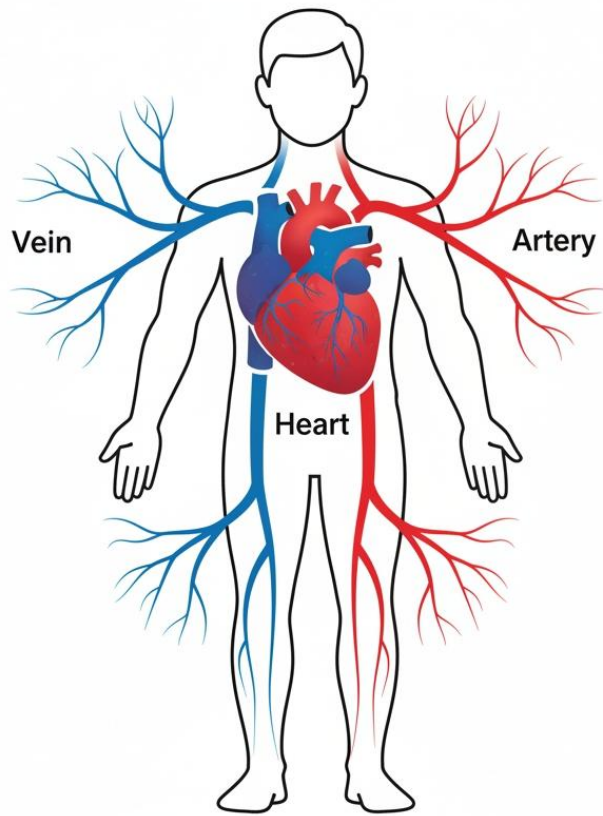
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Identify the main organs of the circulatory system (Heart, Blood vessels).
2. State three functions of blood.
3. Describe simply how blood circulates.
4. Mention ways to keep the heart healthy.

CONTENT

1. The Heart and Blood Vessels The circulatory system is like a delivery network inside your body. The heart is the pump, blood vessels are the roads, and blood is the delivery truck.

- **The Heart:** It is a strong muscle about the size of your fist, located a little to the left in your chest. Its job is to pump blood non-stop throughout your life. When it pumps, it goes lub-dub, lub-dub – that's your heartbeat.
- **Blood Vessels:** These are tubes that carry blood everywhere.
 - **Arteries:** Thick-walled tubes that carry blood away from the heart to the rest of the body. This blood is rich in oxygen and is bright red. The largest artery is called the aorta.
 - **Veins:** Thinner-walled tubes that carry blood back to the heart from the body. This blood has less oxygen and is darker red. Veins have valves to stop blood from flowing backwards.
 - **Capillaries:** These are the tiniest, thinnest vessels. They connect arteries to veins. They are so thin that oxygen and food can pass through their walls to reach every single cell in your body.

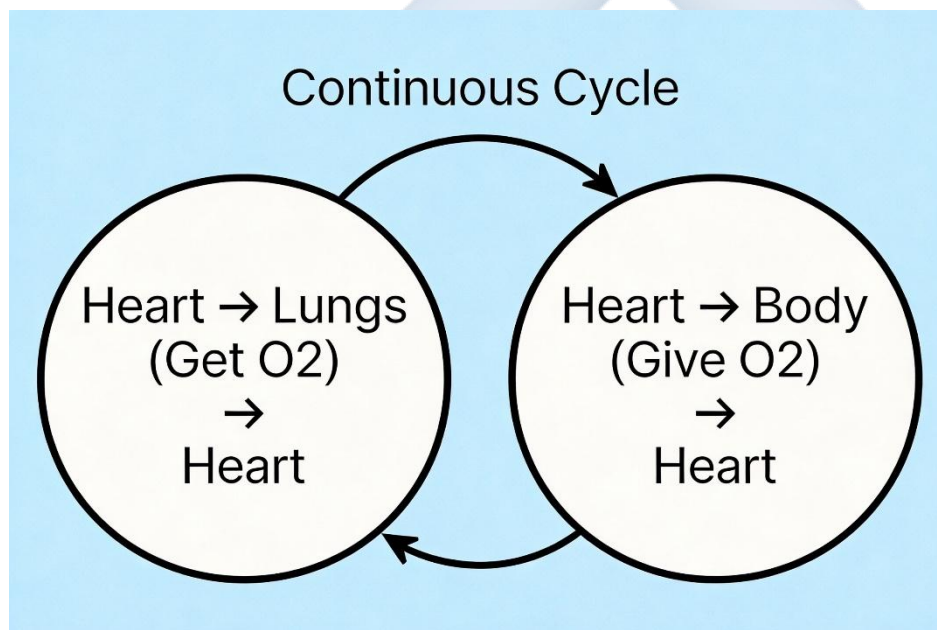


2. Functions of Blood – The Delivery Truck's Jobs Blood is a red liquid that has many important parts doing different jobs:

- **Red Blood Cells:** Shaped like little doughnuts without the hole. They contain hemoglobin which carries oxygen from your lungs to all parts of your body. They also carry some carbon dioxide waste back to the lungs.
- **White Blood Cells:** They are the body's soldiers or police. They fight germs (like bacteria and viruses) that cause infections and diseases.
- **Platelets:** These are tiny pieces of cells that help to clot blood. When you cut yourself, they stick together like a plug to stop the bleeding and form a scab.
- **Plasma:** This is the yellowish liquid part of blood. It carries digested food (nutrients like sugar and vitamins) from your intestines to your cells, and carries waste products to the kidneys to be removed as urine.

3. The Two Loops of Circulation Blood doesn't just go out and back in one loop. It travels in two connected loops:

- **Loop 1: Heart to Lungs and Back (To get Oxygen)** i. The heart pumps "used" blood (low in oxygen) to the lungs. ii. In the lungs, the blood picks up fresh oxygen and releases carbon dioxide (which you breathe out). iii. The now oxygen-rich blood returns to the heart.
- **Loop 2: Heart to Body and Back (To deliver Oxygen)** i. The heart pumps the oxygen-rich blood out through the arteries to every part of your body (head, toes, fingers, stomach). ii. In the body capillaries, blood drops off oxygen and food to the cells and picks up carbon dioxide and other wastes. iii. The now oxygen-poor blood returns to the heart through the veins, ready to go to the lungs again.



4. Keeping the Heart Healthy Our heart works hard for us, so we must take care of it.

- **Exercise regularly:** Activities like running, skipping, playing football, and dancing make the heart muscle strong.
- **Eat heart-healthy foods:** Eat plenty of fruits (oranges, watermelon), vegetables (ugu, waterleaf), and whole grains. Drink plenty of water. Try to eat less fried food, salty snacks, and sugary drinks.

- **Get enough rest and sleep.**
- **Avoid bad habits:** Do not smoke. Smoking damages the heart and blood vessels.

5. Blood Groups There are four main blood types: A, B, AB, and O. Knowing your blood group is important. If someone loses a lot of blood in an accident, doctors must give them blood from a matching donor. Type O is called the "universal donor" because it can be given to people with other blood types in emergencies.

EVALUATION

1. What is the main job of the heart?
2. Which blood vessels carry blood away from the heart?
3. What is the job of the White Blood Cells in our blood?
4. Why is exercise good for our heart?
5. What happens to blood when it goes to the lungs?
6. Name two of the four main blood groups.

ASSIGNMENT

1. **(Practical):** Place two fingers gently on the side of your neck or on your wrist. Feel your pulse (heartbeat). Count how many times it beats in 30 seconds. Multiply by 2 to get your beats per minute. Write it down.
2. **(Drawing):** Draw a simple love-heart shape and color it red. Label it "My Heart".
3. **(Written):** List two things you can do to keep your heart healthy and two things that are bad for your heart.

WEEK 5: MIDTERM EXAMINATION

(Assessment of topics covered in Weeks 1-4)

- **Note for the Teacher:** The examination should include questions from all topics covered so far: Living Things & Environment, Plant Structure, Animal Classification, and the Human Circulatory System. Questions should test understanding, application, and simple reasoning, using familiar, local examples where possible.



WEEK 6: MIDTERM BREAK

(Rest and relaxation period)



WEEK 7: HUMAN BODY: THE DIGESTIVE SYSTEM

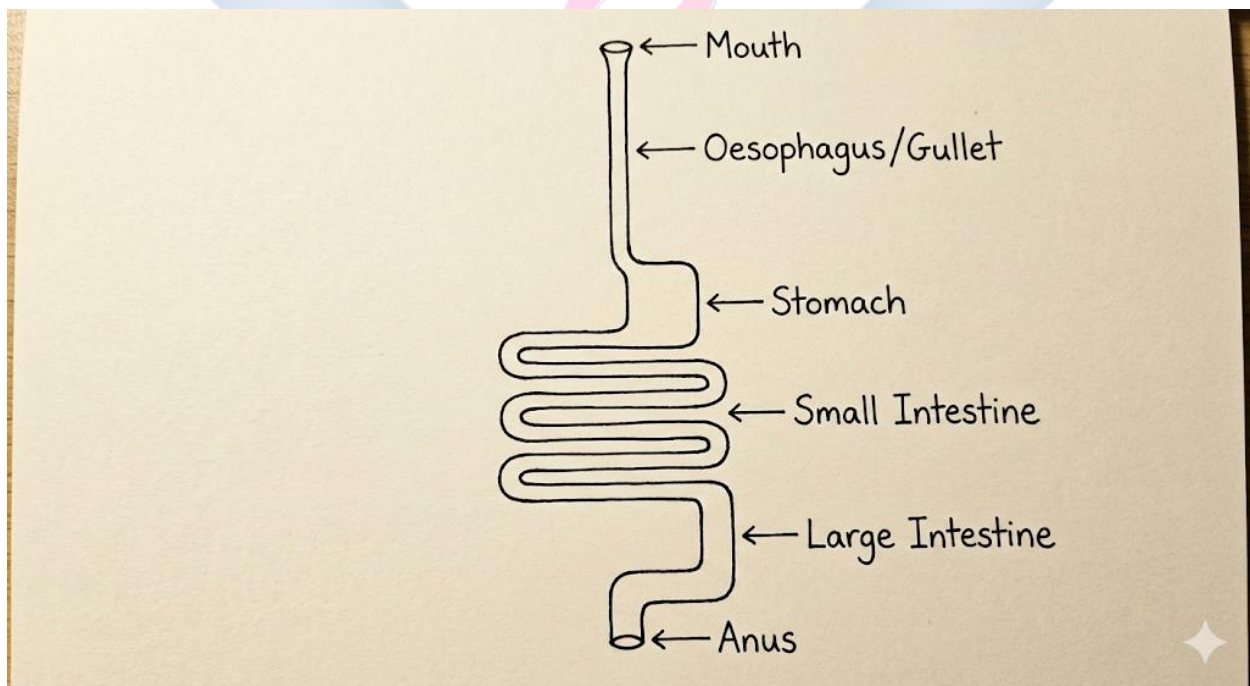
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define digestion.
2. Identify the main parts of the alimentary canal (Mouth, Gullet, Stomach, Intestines).
3. Explain what happens to food in the stomach and mouth.
4. State two habits for good digestive health.

CONTENT

1. What is Digestion? Digestion is the process by which our body breaks down the food we eat into tiny, tiny pieces that are small enough to be absorbed into our blood and used for energy, growth, and repair. Think of it like this: you cannot put a whole yam into a generator to get light. You must cut it and cook it first. Digestion "cuts" and "cooks" our food for the body to use.

2. The Alimentary Canal: The Food Tube This is a long, winding tube that runs from your mouth to your anus. Food travels through it. Each part has a special job.



- **Mouth:** Where digestion begins. You put food here.
 - **Teeth** chop and grind food into smaller bits (mechanical digestion).

- **Saliva (spit)** is a watery juice that wets the food, making it easy to swallow. It also contains an enzyme that starts breaking down starchy foods like rice and garri.
- **Gullet (Oesophagus):** This is a muscular pipe that connects your mouth to your stomach. It does not digest food. Its job is to push the soft lump of chewed food (called a bolus) down to your stomach using wave-like muscle squeezes called peristalsis. This is why you can swallow water even while standing on your head!
- **Stomach:** A stretchy, J-shaped bag that acts like a mixer and blender.
 - It churns the food around.
 - It produces very strong gastric juices (acid and enzymes) that break down proteins (like the meat in your soup).
 - The food becomes a thick, soupy liquid called chyme.
- **Small Intestine:** This is a very long (about 6 meters!), coiled tube where most digestion and absorption happens.
 - Juices from the liver (bile, which breaks down fats) and pancreas (enzymes for all food types) come here.
 - The inner walls of the small intestine are covered in tiny finger-like projections called villi. These villi have two jobs:
 - i. They produce juices to finish digesting the food.
 - ii. They absorb the digested nutrients (sugars, amino acids, vitamins, fats) directly into the bloodstream.
- **Large Intestine:** This is a wider, shorter tube.
 - Its main job is to absorb water and minerals from the leftover liquid waste.
 - The waste becomes more solid, forming faeces (poo).
- **Anus:** This is the opening at the end of the large intestine where solid waste leaves the body.

3. The Journey of a Meal (Example: Eba and Soup) Let's follow a bite of eba and soup: a.

Mouth: Teeth chew the eba. Saliva mixes with it. b. **Gullet:** The chewed lump is swallowed and pushed down. c. **Stomach:** Strong juices break down the fish/meat from the soup. Everything is

churned into a soup mixture. d. **Small Intestine:** Juices from the liver and pancreas work on the oil (fat) in the soup. Nutrients from the eba (carbohydrate), meat (protein), and vegetables (vitamins) are absorbed through the villi into the blood. e. **Large Intestine:** Water is taken back. Leftover fibre from the vegetables and other wastes form poo. f. **Anus:** Poo is passed out when you go to the toilet.

4. Digestive Health To keep your digestive system happy and working well:

- **Chew your food well:** Don't swallow big chunks. Chewing makes the stomach's job easier.
- **Drink plenty of water:** This helps food move smoothly and prevents constipation (hard, difficult poo).
- **Eat foods rich in fibre:** Fibre (found in fruits like oranges, vegetables like ugu, and whole grains) adds bulk to waste and helps it pass easily.
- **Wash your hands before eating:** This prevents harmful germs from entering your mouth and causing stomach upset.
- **Eat at regular times:** Don't skip meals.

EVALUATION

1. What is the simple meaning of digestion?
2. Name the part of the alimentary canal where: a) Digestion begins. b) Water is absorbed.
3. What are the two main jobs of the teeth and saliva in the mouth?
4. What happens to food in the stomach?
5. Why is the small intestine so important?
6. Mention two good habits that can help you avoid a stomach ache.

ASSIGNMENT

1. **(Drawing):** Draw the outline of the human digestive system. Label these three parts: Stomach, Small Intestine, Large Intestine.
2. **(Written - Sequencing):** Rearrange these parts in the correct order that food passes through them: Stomach, Mouth, Large Intestine, Gullet, Small Intestine.

3. **(Oral/Practical):** Talk to a family member. Find out one traditional home remedy in your culture for treating a simple stomach upset (e.g., ginger tea, unripe pawpaw). Share it with the class.



WEEK 8: HUMAN BODY: THE SKELETAL SYSTEM

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define the skeletal system.
2. List three functions of the skeleton.
3. Identify major bones (Skull, Ribs, Spine, Femur).
4. Explain what a joint is.

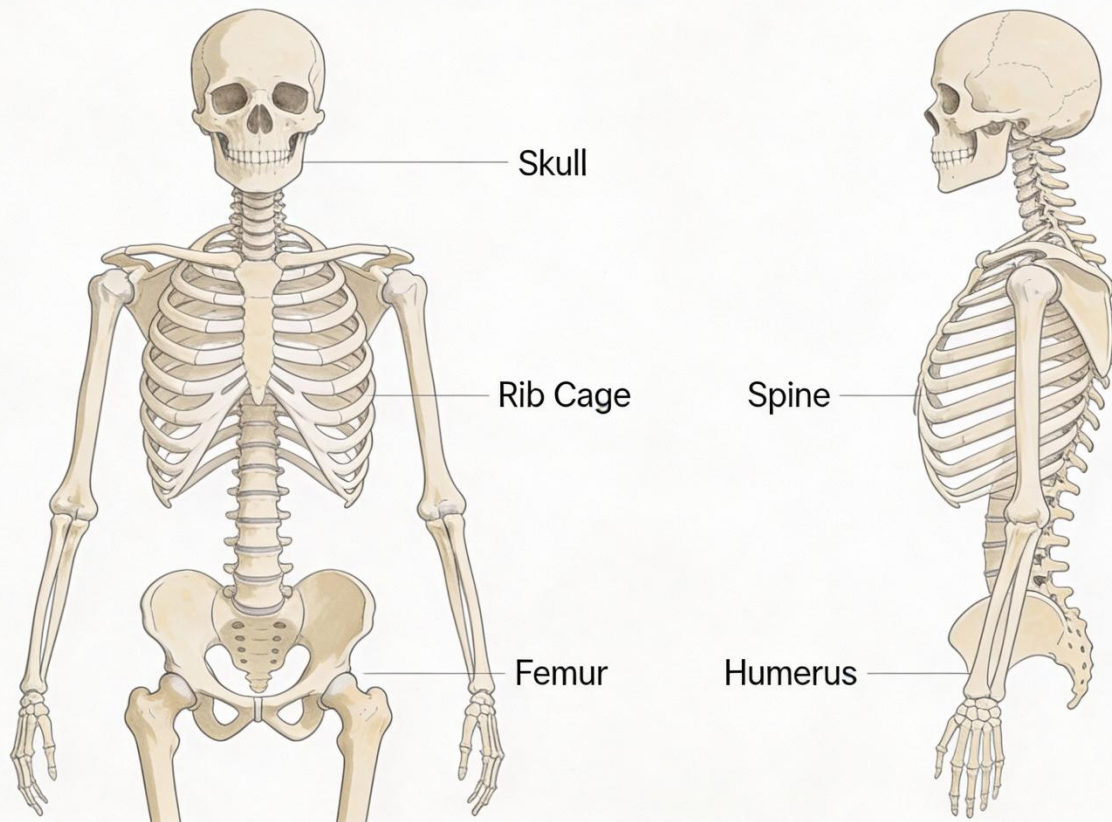
CONTENT

1. What is the Skeletal System?

The skeletal system is all the bones in your body joined together. It is your body's framework or scaffolding. An adult has 206 bones. Without bones, your body would be like a sack of water—you couldn't stand, walk, or hold your shape. Bones are hard and strong because they contain minerals like calcium.

2. Major Bones and Their Jobs

- **Skull:** This is the bone of your head. It is like a hard helmet that protects your delicate brain from injury.
- **Rib Cage:** This is a cage made of curved bones (ribs) around your chest. It acts like a shield, protecting your heart and lungs.
- **Backbone (Spine or Vertebral Column):** This is a stack of 33 small bones called vertebrae. It runs down the centre of your back.
 - It holds you upright.
 - It protects the spinal cord—the main bundle of nerves that carries messages between your brain and body.
- **Femur:** This is your thigh bone. It is the longest and strongest bone in your body.
- **Other important bones:**
 - **Pelvis:** A bowl-shaped bone that supports your intestines and, in women, protects a baby before birth.
 - **Humerus:** The bone in your upper arm.



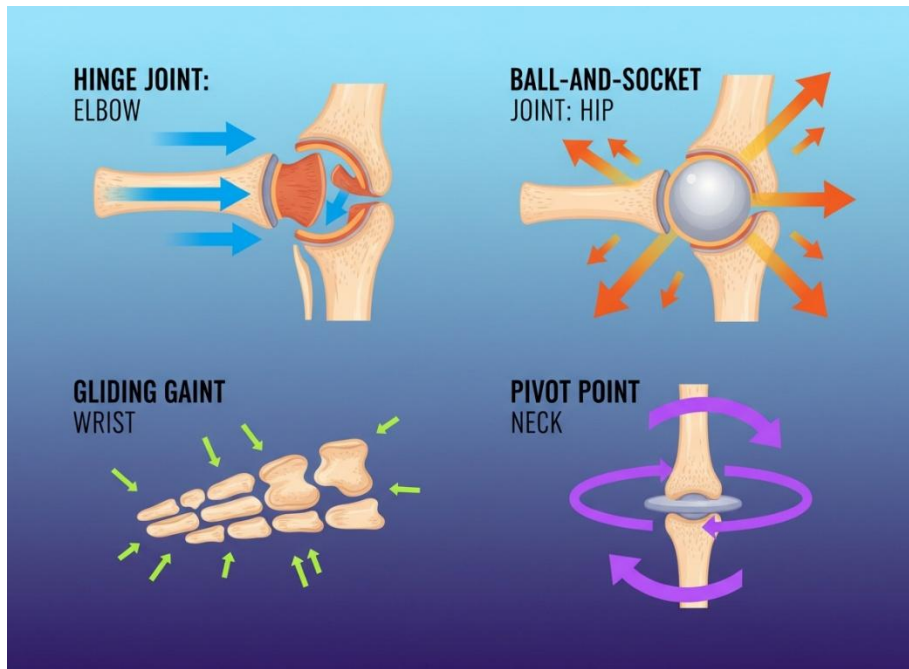
3. Three Main Functions of the Skeleton

a. **Support and Shape:** Just like the wooden frame of a house gives it shape, your skeleton gives your body its shape and holds you up against gravity. b. **Protection:** Bones form a protective box around soft, delicate organs. The skull protects the brain. The rib cage protects the heart and lungs. The spine protects the spinal cord. c. **Movement:** Bones cannot move by themselves. They work with muscles. Muscles are attached to bones by tough cords called tendons. When a muscle contracts (gets shorter and fatter), it pulls on the bone, causing movement (like bending your arm).

4. Joints: Where Bones Meet A joint is the place where two or more bones meet. Joints allow us to bend, twist, and move.

- **Hinge Joint:** Allows back-and-forth movement like a door hinge. Examples: Elbow and Knee.
- **Ball-and-Socket Joint:** Allows movement in almost all directions (round and round). Examples: Shoulder and Hip joints.

- **Gliding Joint:** Allows bones to slide over one another. Example: Bones in your wrist and ankle.
- **Pivot Joint:** Allows rotating/twisting movement. Example: The joint in your neck that lets you turn your head side to side.



5. Bone Care

- **Eat calcium-rich foods:** Calcium makes bones hard and strong. Drink milk, eat yogurt, cheese, ugu, and crayfish.
- **Get sunlight and Vitamin D:** Vitamin D (from sunlight on our skin and from eggs/fish) helps your body use calcium.
- **Exercise:** Activities like running and jumping make bones stronger.
- **Be safe:** Be careful during play to avoid fractures (broken bones).

EVALUATION

1. What is the main job of the skeleton?
2. Which bone acts like a helmet for the brain?
3. Why is the rib cage important?
4. Which is the longest bone in your body? Where is it found?

5. What is a joint? Name the type of joint in your knee.
6. Why should children drink milk?

ASSIGNMENT

1. **(Practical/Observation):** Gently feel your knee cap. Is it hard or soft? Write down what you feel. Now, feel your ear. Is it hard or soft? The hard part is bone, the soft part is cartilage (which is also part of the skeletal system!).
2. **(Drawing):** Draw a simple stick-figure skeleton. Label the Skull and the Rib Cage.
3. **(Written):** List three different places in your body where you have movable joints (e.g., Wrist, Ankle, Jaw).



WEEK 9: MATTER AND MATERIALS

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define matter.
2. List the three states of matter (Solid, Liquid, Gas).
3. Mention properties of each state.
4. Describe how matter changes state (e.g., Ice to Water).

CONTENT

1. What is Matter? Matter is anything that has mass (we can feel its weight) and takes up space (volume). Everything you can see, touch, smell, and sometimes not see, is made of matter.

- **Examples you can see and touch:** Your desk, your book, your water bottle, your shoe.
- **Examples you cannot see but is matter:** The air around you. Air has weight and takes up space (you can feel it when the wind blows).

2. The Three States of Matter Matter exists in three main forms: Solid, Liquid, and Gas. The state depends on how close the tiny particles that make up the matter are and how much they move.

STATES OF MATTER: NIGERIAN EXAMPLES		
STATE	PICTURE EXAMPLE	PROPERTIES
SOLID		• Fixed shape • Fixed size • Does not flow
LIQUID		• No fixed shape • Fixed volume • Flows
GAS		• No fixed shape • No fixed volume • Sprads out

Solids:

- Have a fixed shape and a fixed size. A chalk is always chalk-shaped.
- The particles are packed tightly together in a fixed pattern.
- Examples: Wood, iron, plastic chair, school bell.

• Liquids:

- Have no fixed shape. They take the shape of their container.
- Have a fixed volume. One litre of water is always one litre, whether in a bottle or a bowl.
- They flow and can be poured.
- The particles are close but can slide past each other.
- Examples: Water, palm oil, petrol, kerosene, blood, honey.

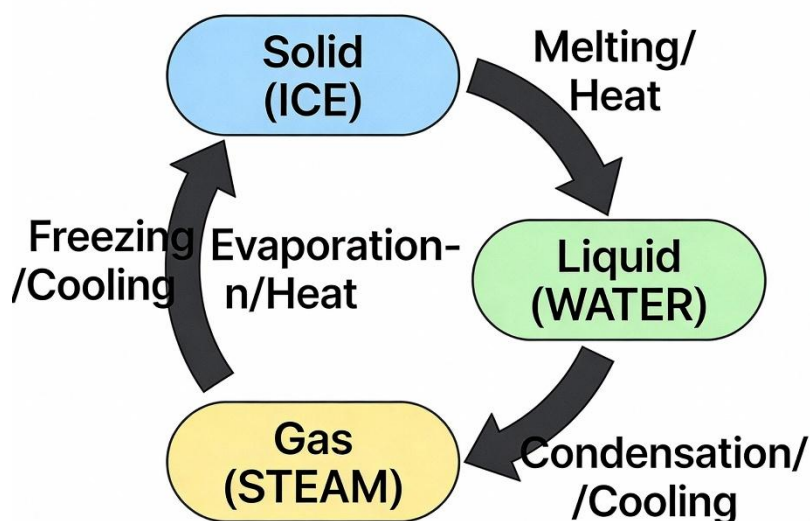
• Gases:

- Have no fixed shape and no fixed volume. They will fill any container completely.
- They spread out (diffuse) easily and quickly.
- The particles are far apart and move very fast.
- Examples: Air (a mixture of gases), the smell of perfume spreading, cooking gas (LPG), carbon dioxide in a soft drink, oxygen.

3. Changing States Matter can change from one state to another when we add heat (warming) or remove heat (cooling). This is called a change of state.

- **Melting:** Solid → Liquid. Adding heat makes particles move faster and break free from their fixed positions. Example: Ice block (solid) melts to form water (liquid) on a hot day.
- **Freezing:** Liquid → Solid. Removing heat makes particles slow down and lock into place. Example: Water (liquid) put in a freezer becomes ice (solid).
- **Evaporation:** Liquid → Gas. Adding heat makes liquid particles move so fast they escape into the air as gas/vapour. Example: Water in a pot boils away as steam. Wet clothes dry in the sun.

- **Condensation:** Gas → Liquid. Cooling a gas makes its particles slow down and come closer together to form a liquid. Example: Water droplets forming on the outside of a cold bottle of water on a hot day. Clouds forming in the sky.



EVALUATION

1. Your school bag is matter. Give two reasons why.
2. Name the three states of matter.
3. Why can you pour garri from a cup into a bowl? What state of matter is garri?
4. Air has no fixed shape. What state of matter is it?
5. What change of state happens when you put a pot of water on fire to boil?
6. What do we call the process where water vapour turns back to liquid on a cold surface?

ASSIGNMENT

1. **(Written):** Look around your classroom. List three objects that are solids.
2. **(Written/Practical):** Go to your kitchen at home. Name two liquids used for cooking.
3. **(Drawing):** Draw a simple sequence of three pictures: 1. An ice cube. 2. The ice cube melting into water. 3. The water evaporating as wavy lines (steam) above it.

WEEK 10: MIXTURES AND SEPARATION

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define a mixture.
2. Differentiate between a mixture and a pure substance.
3. Describe three methods of separating mixtures (Filtration, Evaporation, Sedimentation).
4. Identify practical applications of separation.

CONTENT

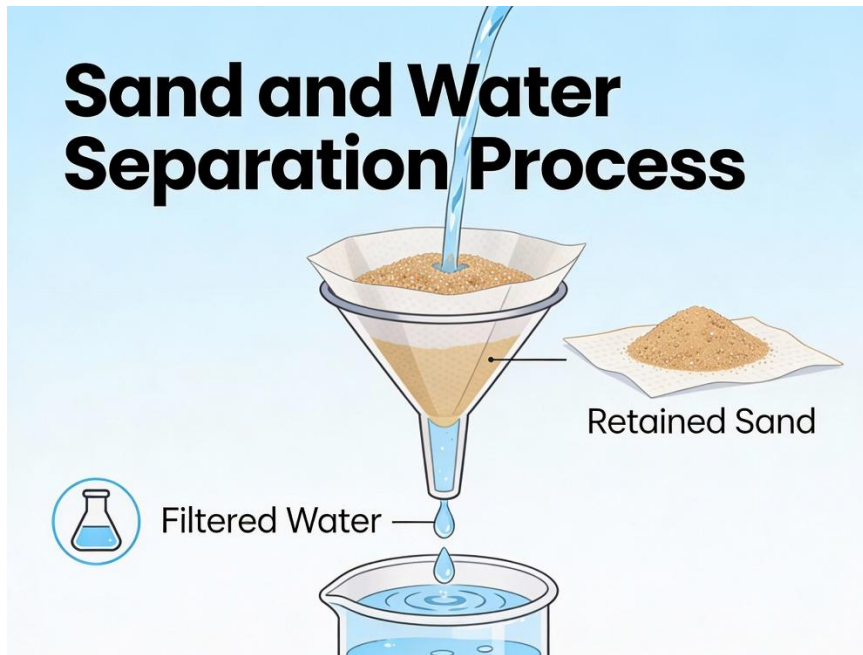
1. What is a Mixture? A mixture is formed when two or more different substances are put together without joining chemically. Each substance keeps its own properties. The parts of a mixture can usually be separated by physical means.

- **Examples:** Sand and salt mixed together. Rice and beans. Garri and sugar. Water and groundnut oil.
- **Pure Substance:** A material made up of only one type of particle. It has the same properties throughout. Examples: Pure water (H₂O only), pure salt, pure sugar, 24-carat gold.

2. Common Methods of Separating Mixtures We use different methods depending on the properties of the substances in the mixture.

- **Filtration/Sieving:** Used to separate a solid from a liquid or solids of different sizes.
 - **How it works:** A filter (like a sieve, cloth, or filter paper) has tiny holes. The liquid or smaller particles pass through, but larger solid particles are trapped.
 - **Nigerian Examples:**
 - Separating tea leaves from tea using a sieve.
 - Separating sand from well water using a cloth filter.
 - A woman using a local basket (*) to separate stones from beans or rice.

Sand and Water Separation Process



Evaporation: Used to separate a solid dissolved in a liquid (like salt in water).

- **How it works:** The mixture is heated. The liquid turns to gas and escapes into the air, leaving the solid behind.
- **Nigerian Examples:**
 - Getting salt from sea water in places like Badagry. They channel seawater into shallow ponds and let the sun evaporate the water.
 - Making local salt from salt springs.
- **Sedimentation and Decantation:** Used to separate a heavy, insoluble solid from a liquid (like mud from water).
 - **Step 1 - Sedimentation:** The mixture is left to stand undisturbed. The heavier solid particles sink to the bottom (settle).
 - **Step 2 - Decantation:** The clear liquid at the top is carefully poured off into another container.
 - **Nigerian Example:** Fetching muddy river water into a bucket and allowing it to settle before using the clearer water on top.
- **Using a Magnet:** Used to separate magnetic materials (like iron) from non-magnetic materials.
 - **Example:** Using a magnet to remove iron filings or nails from sand.

3. Why is Separation Important? (Practical Applications)

- To make water safe and clean for drinking (filtration, sedimentation).
- To get useful materials (getting salt from seawater).
- In cooking (sieving flour to remove lumps, separating chaff from rice).
- In recycling (using magnets to separate steel cans from other rubbish).

EVALUATION

1. What is a mixture? Give one example from your kitchen.
2. You have a cup of garri with sand in it. Which separation method is best to clean it? Why?
3. How would a farmer separate tiny weed seeds from his harvested beans?
4. What method is used in salt ponds? Explain how it works.
5. After sedimentation, what is the next step to get the clear liquid?
6. Is pure, bottled water a mixture? Yes or No? Explain.

ASSIGNMENT

1. **(Practical):** At home, mix two spoonfuls of sand with a cup of water in a transparent glass. Let it sit for 30 minutes. Observe and write down what happens. (This is sedimentation).
2. **(Written):** List two tools or items used for separating mixtures in your home (e.g., sieve, cloth, magnet).
3. **(Drawing/Written):** Draw a simple comic strip (2 boxes) showing how to separate a mixture of iron nails and sand. Write one sentence explaining your method.

WEEK 11: REVISION

CONTENT

- **Review of Living Things:** Recap Habitats (Aquatic, Terrestrial), Adaptations (fish fins, camel hump), Food Chains (Producer → Consumer), and Interdependence.
- **Review of Plants:** Recap parts (Root, Stem, Leaf, Flower) and their jobs. Revise Photosynthesis (Water + CO₂ + Sunlight → Food + Oxygen).
- **Review of Animals:** Recap Vertebrates (5 groups: Mammals, Birds, Reptiles, Amphibians, Fish) vs. Invertebrates.
- **Review of Human Body Systems:**
 - **Circulatory:** Heart (pump), Blood vessels (arteries, veins), Functions of blood.
 - **Digestive:** Parts (Mouth, Stomach, Intestines), process of digestion.
 - **Skeletal:** Functions (Support, Protection, Movement), major bones, joints.
- **Review of Matter and Mixtures:** Three states (Solid, Liquid, Gas), their properties. Methods of separation (Filtration, Evaporation, Sedimentation).
- **Practice Questions for End of Term:** Use questions from the Evaluation sections of all weeks to create a short revision quiz. Focus on application: "Why does a fish have gills?" "What will happen if you leave ice in the sun?"

PRIMARY 5 BASIC SCIENCE & TECHNOLOGY SCHEME OF WORK

SECOND TERM

Week	Topic	Content
1	Energy and Its Forms	Forms of energy; Energy transformation; Conservation of energy; Renewable and non-renewable energy sources
2	Electricity and Magnetism	Electric circuits - series and parallel; Electrical symbols; Magnetic materials; Uses of magnets; Electromagnets
3	Force and Motion	Types of forces; Effects of force; Friction; Speed and velocity; Measuring motion
4	Work, Energy and Power	Concept of work; Relationship between work and energy; Simple calculations; Efficiency of machines
5	MIDTERM EXAMINATION	Assessment of topics covered in weeks 1-4
6	MIDTERM BREAK	Rest and relaxation period
7	The Earth and Its Resources	Structure of the earth; Natural resources; Minerals and mining; Conservation of resources; Sustainable use
8	Water Cycle and Conservation	The water cycle process; Importance of water; Water treatment; Water pollution; Conservation methods
9	Air Composition and Quality	Gases in the air; Importance of each gas; Air pollution causes and effects; Improving air quality
10	Weather and Climate Change	Weather recording and instruments; Climate zones; Climate change causes; Effects of climate change; What we can do
11	REVISION	Review of all topics covered; Practice exercises; Question and answer sessions
12	EXAMINATION	End of term assessment
13	CLOSING	Report card distribution; Vacation assignment

PRIMARY 5 BASIC SCIENCE & TECHNOLOGY LESSON NOTES

SECOND TERM

WEEK 1: ENERGY AND ITS FORMS

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define energy and list four forms of energy.
2. Explain energy transformation with two everyday examples.
3. State the law of conservation of energy in simple terms.
4. Differentiate between renewable and non-renewable energy sources.

CONTENT

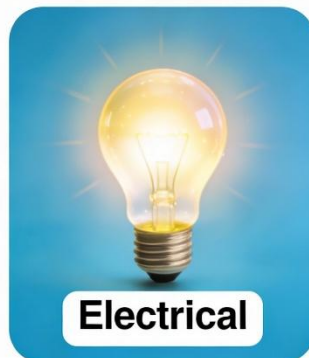
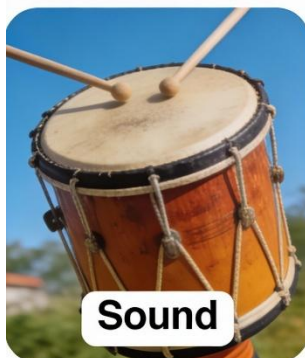
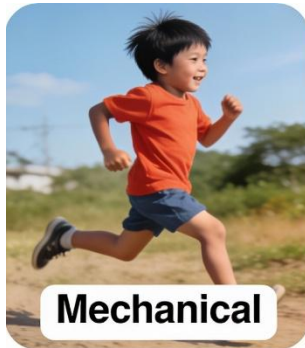
1. What is Energy? Energy is the ability or capacity to do work or make things happen. You cannot see energy itself, but you can see and feel what it does.

- **Without energy:** You could not walk, a car could not move, a phone could not ring, and a bulb could not give light.
- **Simple definition:** Energy is what makes things go, move, or change.

2. Forms of Energy Energy exists in different forms around us.

- **Mechanical Energy:** The energy of movement or position.
 - **Movement (Kinetic):** A rolling football, a spinning fan, a person running.
 - **Position (Potential):** A stone sitting on top of a hill, a stretched rubber band, water held behind a dam.
- **Heat (Thermal) Energy:** The energy that makes things hot. We can feel it.
 - **Examples:** Heat from the sun, from fire, from a hot stove, from our bodies.
- **Light Energy:** The energy that makes things visible. We can see it.
 - **Examples:** Sunlight, light from a bulb, torchlight, candle flame.
- **Sound Energy:** The energy we can hear. It travels through the air as vibrations.
 - **Examples:** Sound from a drum, a ringing bell, someone talking, music from a radio.

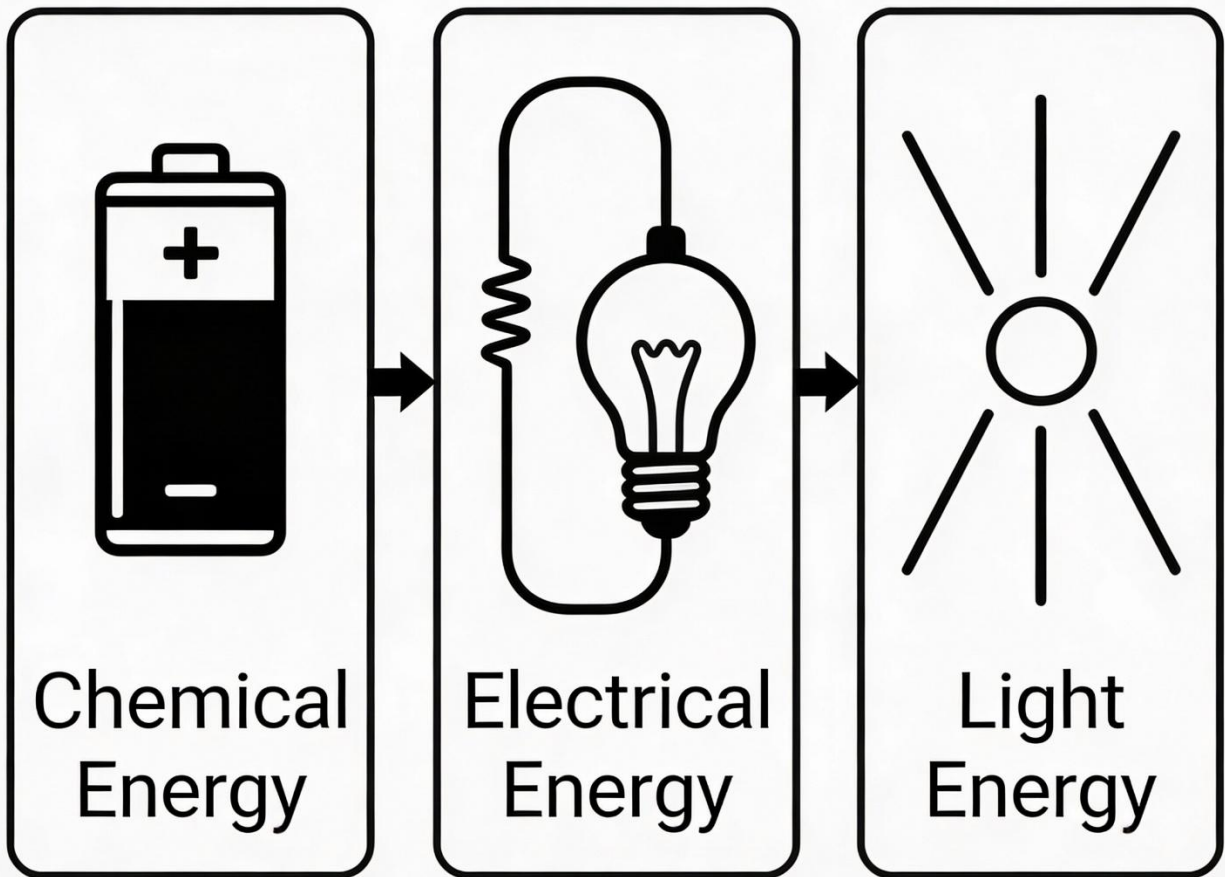
- **Electrical Energy:** The energy carried by electricity. It is very useful in homes.
 - **Examples:** Energy that powers televisions, refrigerators, electric kettles, and street lights.
- **Chemical Energy:** Energy stored inside substances, waiting to be released.
 - **Examples:** Food (gives us energy), petrol/diesel (makes cars move), batteries (power torches and toys), firewood (produces heat when burned).



3. Energy Transformation (Change) Energy is amazing because it can change from one form to another. This is called energy transformation.

- **Example 1: A Battery-Powered Torch**
 - Chemical energy in the battery changes to Electrical energy in the wires.
 - The electrical energy changes to Light energy (and a little Heat energy) in the bulb.
 - **Transformation Path:** Chemical → Electrical → Light & Heat.
- **Example 2: Rubbing Your Palms Together**
 - You use Mechanical energy (movement of your hands).
 - This changes to Heat energy (your palms feel warm).

- **Transformation Path:** Mechanical → Heat.
- **Example 3: A Person Eating and Running**
 - Chemical energy in food is changed by your body.
 - It becomes Mechanical energy in your muscles to run.
 - **Transformation Path:** Chemical (food) → Mechanical (movement).

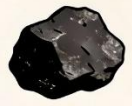


4. Law of Conservation of Energy This is a very important scientific law. It says: "Energy cannot be created or destroyed. It can only be changed from one form to another."

- **Simple Explanation:** The total amount of energy in the universe stays the same. When energy changes form, no energy is lost; it just becomes a different type.
- **Analogy:** It's like pouring water from a bottle into a cup. The water changes container (form), but the amount of water (energy) is the same.

5. Renewable and Non-Renewable Energy Sources

- **Renewable Energy Sources:** These are natural sources that will not run out. They are often clean and can be used again and again.
 - **Solar Energy:** From the sun (solar panels, sunlight).
 - **Wind Energy:** From moving air (windmills).
 - **Water Energy (Hydro):** From flowing water (dams like Kainji Dam).
 - **Biomass:** From plants and animal waste (firewood, biogas).
- **Non-Renewable Energy Sources:** These are sources that can finish (run out) one day because they take millions of years to form. They are often burned to produce energy, which causes pollution.
 - **Crude Oil (Petroleum):** Used to make petrol, diesel, kerosene.
 - **Coal:** Used in some power plants and for heating.
 - **Natural Gas:** Used for cooking and electricity.

Renewable	Non-Renewable
	
	
	

EVALUATION

1. What is energy? Give one example of something that needs energy to work.
2. Name any four forms of energy.
3. What energy transformation takes place when someone uses a generator to power a light bulb? (Hint: Think of the fuel and the result).
4. According to the Law of Conservation, can we create new energy from nothing? (Yes/No).
5. Why is sunlight called a renewable source of energy?
6. Petrol is made from crude oil. Is crude oil a renewable or non-renewable source? Why?

ASSIGNMENT

1. **(Drawing):** Draw and color a large sun. Write two forms of energy it provides next to it (e.g., Light and Heat).
2. **(Written/Practical):** Walk around your home. List three appliances or devices that use electrical energy.
3. **(Written - Explanation):** Trace the energy transformation in a person using a kerosene lantern. Start with the chemical energy in kerosene. What does it transform into?

WEEK 2: ELECTRICITY AND MAGNETISM

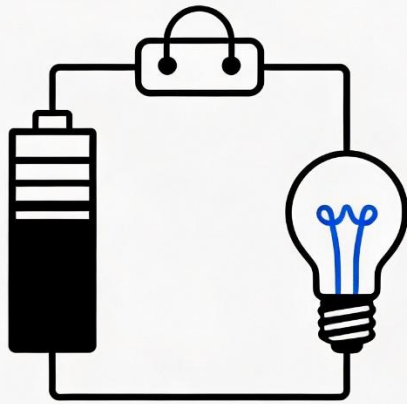
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Describe a simple electric circuit.
2. Differentiate between series and parallel circuits.
3. Identify common electrical symbols.
4. List magnetic materials and uses of magnets.

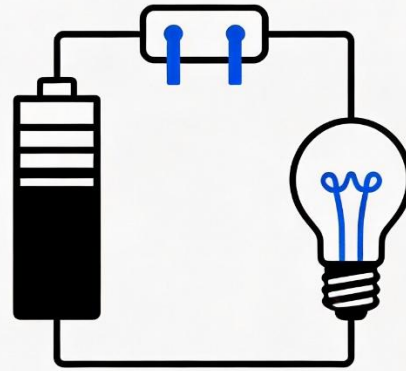
CONTENT

1. Simple Electric Circuit For electricity to flow and do useful work (like light a bulb), it needs a complete, unbroken path. This path is called an electric circuit.

- **Four Essential Parts:**
 - i. **Source:** Provides the electrical energy (e.g., Battery, Generator).
 - ii. **Load:** The device that uses the electricity (e.g., Bulb, Buzzer, Fan).
 - iii. **Wires (Connectors):** Conducting materials (usually copper) that carry electricity from the source to the load and back.
 - iv. **Switch:** A device used to open (break) or close (complete) the circuit, turning the load ON or OFF.
- **Closed vs. Open Circuit:**
 - **Closed Circuit:** The switch is ON. The path is complete. Electricity flows. The bulb lights up.
 - **Open Circuit:** The switch is OFF. There is a gap/break in the path. Electricity cannot flow. The bulb is off.



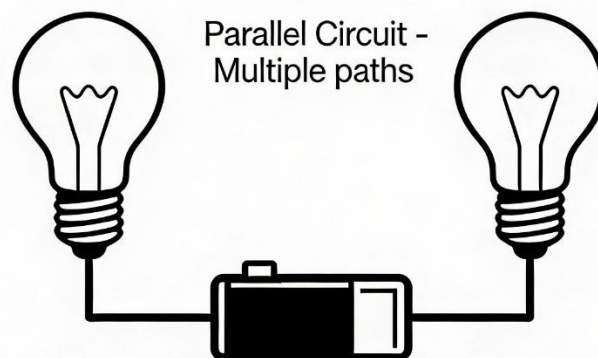
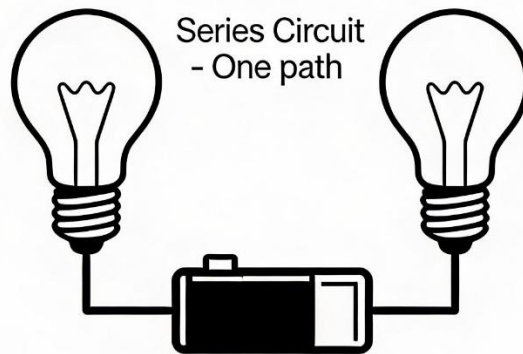
Closed Circuit - Bulb ON



Open Circuit - Bulb OFF

2. Series and Parallel Circuits

- **Series Circuit:** All components (like bulbs) are connected one after the other in a single loop. There is only one path for electricity to flow.
 - **Disadvantage:** If one bulb blows or is removed, the entire path is broken. All other bulbs go OFF.
 - **Example:** Some old Christmas tree lights.
- **Parallel Circuit:** Components are connected on separate branches. There are multiple paths for electricity to flow.
 - **Advantage:** If one bulb blows, the other paths are still complete. The other bulbs stay ON.
 - **Example:** The wiring in our homes, schools, and offices. This is why you can turn off the light in one room without affecting another.



3. Electrical Symbols Scientists and electricians use simple symbols to draw circuits quickly.

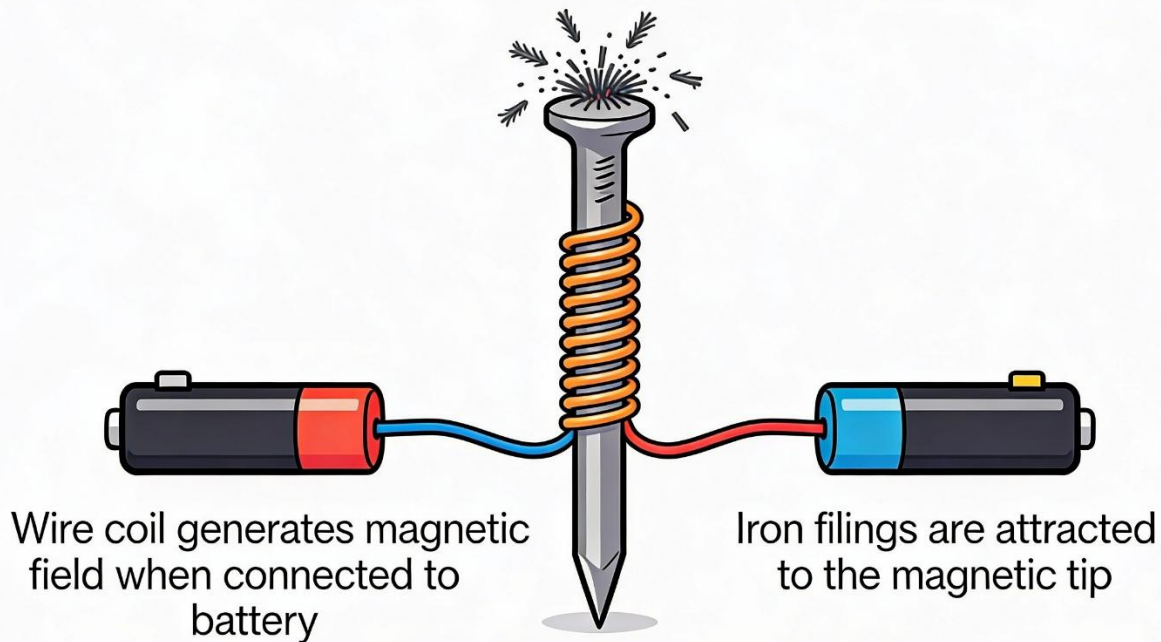
- **Cell/Battery:** A pair of lines (one long +, one short -).
- **Bulb/Lamp:** A circle with a cross inside and a loop.
- **Switch:** A line with a break and a hinged lever.
- **Wire:** A straight line.

Symbol: ☀️	Symbol: 🌧️	Symbol: 🌳
		
Picture: A yellow sun with rays	Picture: Blue raindrops falling	Picture: A green tree with leaves
Name: Sun	Name: Rain	Name: Tree

4. Magnetism and Electromagnets

- **Magnets:** Objects that can attract certain metals (mainly iron and steel). They have a North (N) and South (S) pole.
 - **Magnetic Materials:** Attracted to magnets (e.g., Iron, Steel, Nickel, Cobalt).
 - **Non-Magnetic Materials:** Not attracted to magnets (e.g., Wood, Plastic, Glass, Paper, Aluminium).
- **Uses of Magnets:** In refrigerator door seals, speakers, compasses for direction, magnetic latches on bags/cupboards, and to separate iron scraps from waste.
- **Electromagnet:** A temporary magnet made by coiling a wire around an iron nail (or core) and connecting the wire to a battery.
 - It only works when electric current is flowing through the wire.
 - It becomes stronger with more coils of wire or a bigger battery.
 - **Uses:** In cranes to lift heavy iron scraps, in electric bells, and in telephones.

How to Make an Electromagnet



EVALUATION

1. Name the four parts needed to make a simple electric circuit.
2. What is the main problem with a series circuit?
3. Why are houses wired using parallel circuits instead of series circuits?
4. Draw the symbol for an electric bulb.
5. Which of these will be attracted to a magnet: iron key, plastic ruler, aluminium foil, steel spoon?
6. What two things do you need to make a simple electromagnet?

ASSIGNMENT

1. **(Drawing):** Using the symbols you have learned, draw a circuit diagram with a battery, a closed switch, and two bulbs connected in parallel.

2. **(Practical/Written):** Go home and find three different items that use magnets (e.g., fridge door, purse clasp, toy). List them.
3. **(Written - Explanation):** In your own words, explain why an electromagnet is called a "temporary" magnet.



WEEK 3: FORCE AND MOTION

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define force and mention two types of forces.
2. List three effects of force on an object.
3. Explain the concept of friction.
4. Differentiate between speed and velocity.

CONTENT

1. What is a Force? A force is a push or a pull acting upon an object. Forces can make objects start moving, stop moving, or change their motion.

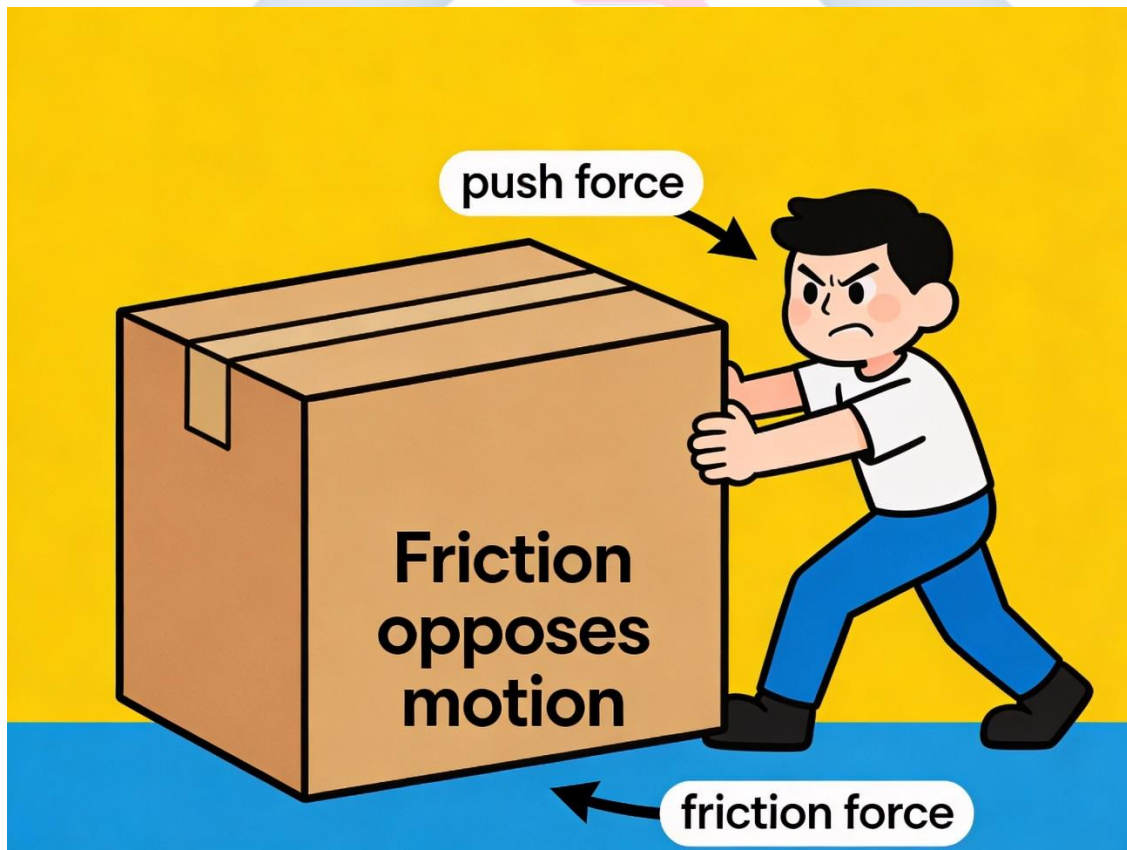
- **Contact Forces:** The objects must touch each other for the force to act.
 - **Examples:** Pushing a door open, kicking a ball, writing with a pencil, pulling a drawer.
- **Non-Contact (Action-at-a-Distance) Forces:** The objects do not need to touch. The force acts through space.
 - **Gravitational Force:** The force that pulls everything towards the centre of the Earth. It gives us weight and makes things fall down.
 - **Magnetic Force:** The force a magnet uses to pull iron objects without touching them.
 - **Electrostatic Force:** The force between charged objects (e.g., a plastic ruler rubbed on hair can attract small pieces of paper).

2. Effects of Force Force can do several things to an object:

a. **It can start motion:** A stationary football begins to move when kicked. b. **It can stop motion:** A goalkeeper uses his hands to stop a moving ball. c. **It can change speed:** Pedaling a bicycle faster makes it speed up; applying brakes makes it slow down. d. **It can change direction:** A tennis player uses a racket to change the direction of the ball. e. **It can change shape:** Squeezing a sponge, stretching a rubber band, or bending a ruler.

3. Friction: The Force That Opposes Motion Friction is a force that acts between two surfaces that are touching. It opposes (works against) the motion of one surface over the other.

- **Causes of Friction:** No surface is perfectly smooth. Microscopic bumps on surfaces catch on each other, creating resistance.
- **Useful Friction (Good):**
 - Allows us to walk without slipping.
 - Allows car tyres to grip the road for moving and braking.
 - Allows us to hold a pencil or a cup.
- **Harmful Friction (Bad):**
 - Causes wear and tear on shoe soles, machine parts, and tyres.
 - Wastes energy by producing unwanted heat (makes machines hot).
- **Reducing Friction:** We use lubricants (oil, grease) between moving parts, make surfaces smoother (polishing), or use rollers/ball bearings.



4. Speed and Velocity

- **Speed:** This tells us how fast an object is moving. It is the distance covered in a specific time.
 - **Formula:** $\text{Speed} = \text{Distance} \div \text{Time}$ (e.g., km/h, m/s).
 - **Example:** "The car's speed was 60 kilometres per hour."
- **Velocity:** This tells us how fast AND in what direction an object is moving. Speed + Direction = Velocity.
 - **Example:** "The car's velocity was 60 km/h heading North."
 - A change in speed OR direction means a change in velocity. Driving around a roundabout at constant speed means your velocity is changing because the direction is changing.

EVALUATION

1. Define force. Is gravity a contact or non-contact force?
2. Give one example of a force changing the shape of an object.
3. List three useful effects of friction in our daily lives.
4. Why do mechanics pour oil into the engines of cars and generators?
5. A car travels from Lagos to Ibadan (a distance of 150 km) in 2 hours. What is its average speed?
6. What is the key difference between speed and velocity?

ASSIGNMENT

1. **(Practical/Observation):** Try to push a book on a rough carpet. Then push the same book on a smooth table. On which surface was it easier to push? Write down your observation and state the reason.
2. **(Written):** List two methods used to reduce friction in common machines (e.g., bicycle, door hinge).
3. **(Calculation):** A cyclist covers 30 metres in 10 seconds. Calculate his speed in metres per second (m/s).

WEEK 4: WORK, ENERGY AND POWER

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define work in scientific terms.
2. State the relationship between work and energy.
3. Perform simple calculations for work.
4. Explain the efficiency of machines.

CONTENT

1. Scientific Meaning of "Work"

In everyday language, "work" can mean many things (schoolwork, housework). But in science, Work has a specific meaning: Work is said to be done only when a force causes an object to move in the direction of the force.

- **If there is no movement, NO scientific work is done, even if you feel tired.**
 - **Example:** A man pushing hard against a heavy wall that does not move is doing zero work scientifically.
- **If there is movement in the direction of the force, WORK is done.**
 - **Example:** A woman lifting a bucket of water from the ground upwards is doing work. The force (lifting) causes upward movement.
- **Formula:** $\text{Work (J)} = \text{Force (N)} \times \text{Distance moved in direction of force (m)}$
- **Unit:** The unit of work is the Joule (J). A smaller unit is the kilojoule (kJ).

2. Relationship Between Work and Energy

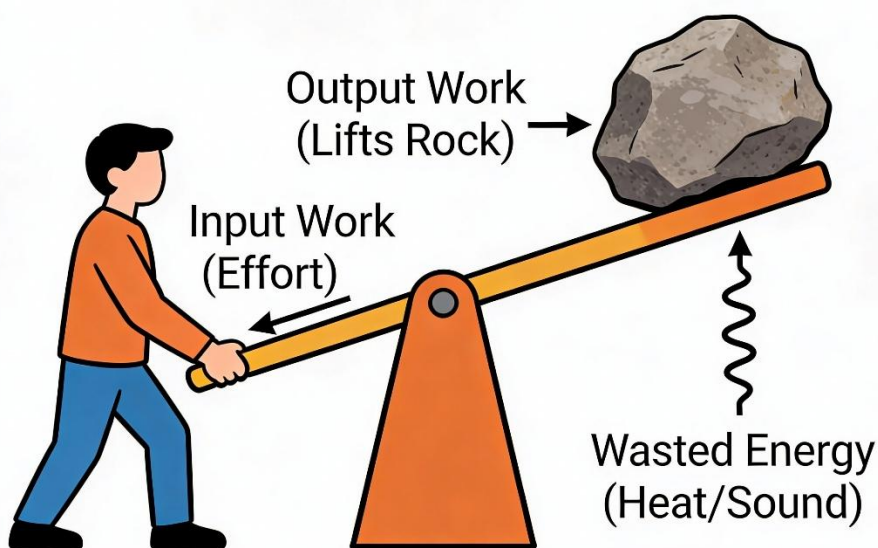
- **Energy is the ability to do work.** You must have energy to be able to do work.
- **Doing work is a way of transferring energy from one object to another.**
- **Example:** When you lift a book, you do work. Your body uses chemical energy from food to provide the force. This work transfers energy to the book, giving it potential energy (energy due to height).

3. Simple Calculations

- **Example 1:** If you use a force of 20 Newtons to push a box 5 metres across the floor.
 - **Work Done** = $20\text{ N} \times 5\text{ m} = 100\text{ Joules (J)}$.
- **Example 2:** A pupil carries a school bag weighing 30 Newtons on his head while walking 100 metres on a flat road.
 - **Is he doing scientific work?** No. The force (upwards, holding the bag) is not causing movement in its direction. The movement (walking forward) is sideways. Therefore, **Work Done = 0 J**.

4. Efficiency of Machines A machine is any device that makes work easier (e.g., lever, pulley, screwdriver, car).

- **Input Work (Energy):** The work/energy you put into the machine.
- **Output Work (Energy):** The useful work/energy you get out of the machine.
- **Efficiency:** This measures how good a machine is at converting input work into useful output work. Some energy is always "wasted" as heat or sound (due to friction), so no machine is 100% efficient.
- We improve efficiency by reducing friction (oiling, greasing) and making machines with smoother parts.



EVALUATION

1. In science, when is work said to be done?
2. A boy sits on a chair reading a book for one hour. Has he done any scientific work? Why or why not?
3. What is the unit for measuring work?
4. Calculate the work done when a force of 50 N moves an object 3 m.
5. How are work and energy related? (Complete the sentence: You need _____ to do work).
6. Why can a machine never be 100% efficient?

ASSIGNMENT

1. **(Calculation):** A woman pulls a cart with a force of 80 Newtons for a distance of 15 metres. Calculate the work done.
2. **(Written):** List two simple machines found in your home (e.g., scissors, bottle opener) and state what work they make easier.
3. **(Written - Explanation):** Explain why a bicycle chain needs oil. How does this relate to the idea of efficiency?

WEEK 5: MIDTERM EXAMINATION

(Assessment of topics covered in Weeks 1-4)



WEEK 6: MIDTERM BREAK

(Rest and relaxation period)



WEEK 7: THE EARTH AND ITS RESOURCES

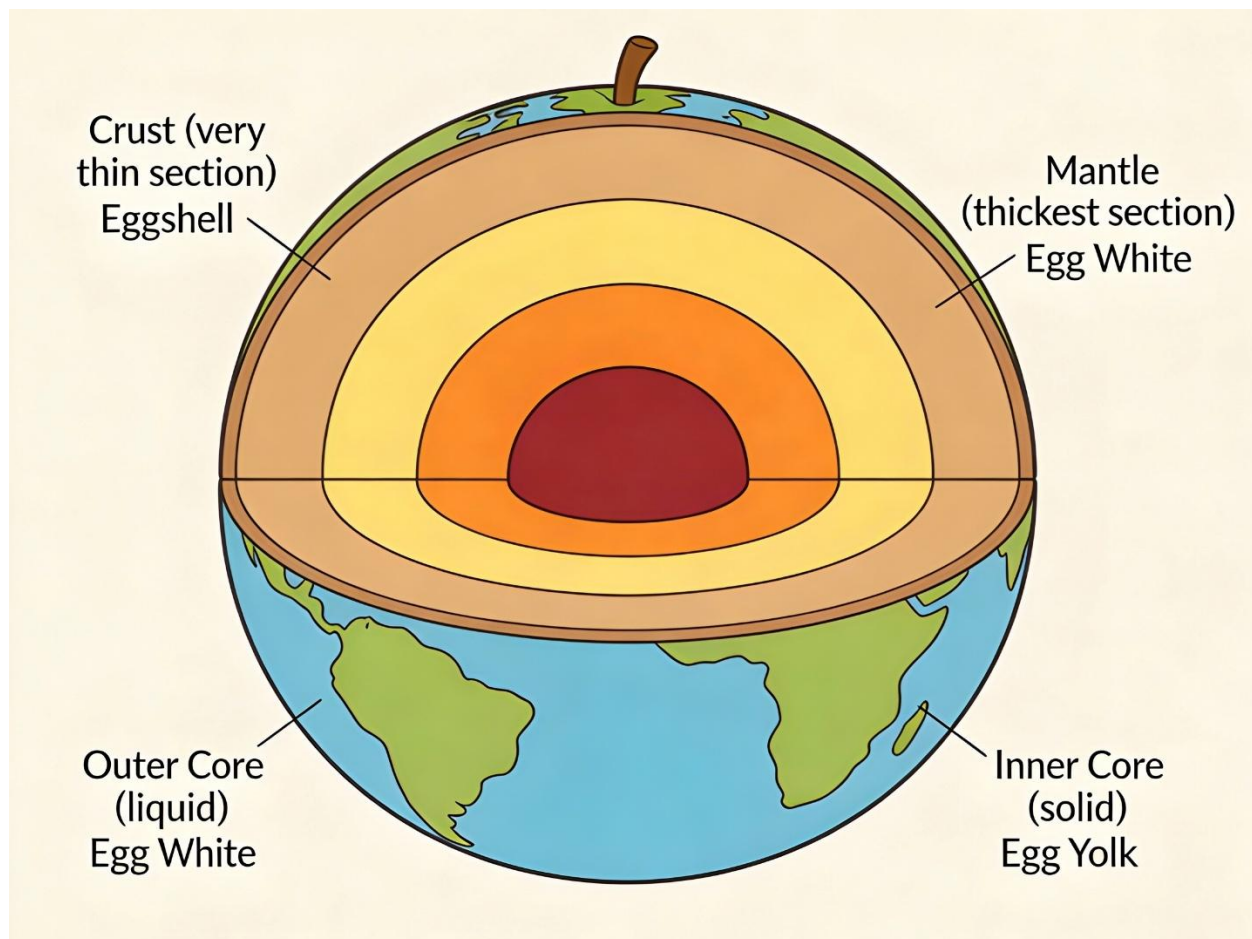
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Describe the structure of the Earth (Crust, Mantle, Core).
2. List examples of natural resources.
3. Explain the difference between minerals and mining.
4. Discuss conservation and sustainable use of resources.

CONTENT

1. Structure of the Earth The Earth is not a solid ball of rock. It is made up of different layers, like an onion or a boiled egg.

- **Crust:** This is the thin, outer layer where we live. It is like the shell of an egg or the skin of an apple. It is made of solid rock and soil. It contains all the land (continents) and ocean floors. It is the thinnest layer.
- **Mantle:** This is the thick, middle layer beneath the crust. It is like the egg white. It is made of hot, solid rock that can flow very slowly over millions of years. The heat and movement here can cause earthquakes and volcanoes.
- **Core:** This is the center of the Earth. It is like the egg yolk. It has two parts:
 - **Outer Core:** Made of hot, liquid metal (mostly iron and nickel).
 - **Inner Core:** Made of solid metal (iron and nickel). It is extremely hot due to great pressure, even though it is solid.



2. Natural Resources Natural Resources are valuable materials and substances that are found in nature and are essential for human life. We use them to make the things we need.

- **Vegetation Resources:** From plants.
 - **Examples:** Trees (for wood, paper, rubber), Crops like cassava and maize (for food), Medicinal herbs, Cotton (for cloth).
- **Water Resources:** From rivers, lakes, and underground.
 - **Examples:** For drinking, farming (irrigation), fishing, transportation (boats), and generating electricity (like at Kainji Dam).
- **Mineral Resources:** Solid, non-living substances dug from the ground.
 - **Examples in Nigeria:** Limestone (used for cement, e.g., in Ewekoro), Tin (mined in Jos), Coal (mined in Enugu), Gold, Iron ore, Crude Oil (a liquid mineral found in the Niger Delta).

- **Land/Soil:** Used for building houses, farming, and animal grazing.

3. Minerals and Mining

- **Minerals:** These are the natural, solid substances themselves (e.g., a piece of gold, a lump of coal).
- **Mining:** This is the process or activity of digging these minerals out of the Earth's crust. It can be done in different ways:
 - **Open-pit Mining:** Digging a big, open hole in the ground (e.g., for limestone).
 - **Underground Mining:** Digging deep tunnels and shafts to reach minerals (e.g., for coal).
 - **Drilling:** For liquid minerals like crude oil.

4. Conservation and Sustainable Use Our natural resources are precious and many are limited.

- **Conservation:** This means protecting and managing our resources wisely so they don't get finished, wasted, or destroyed.
- **Sustainable Use:** This means using resources in a way that meets our needs today without stopping future generations (like your children) from meeting their own needs.
- **How we can conserve:**
 - **Replant Trees (Afforestation):** If we cut one tree for firewood, we should plant two new ones.
 - **Reduce, Reuse, Recycle:** Use less plastic, reuse bottles, recycle paper and cans.
 - **Prevent Pollution:** Don't dump waste into rivers or burn plastics that poison the air.
 - **Use Alternatives:** Use renewable energy (solar) instead of only relying on firewood or petrol.

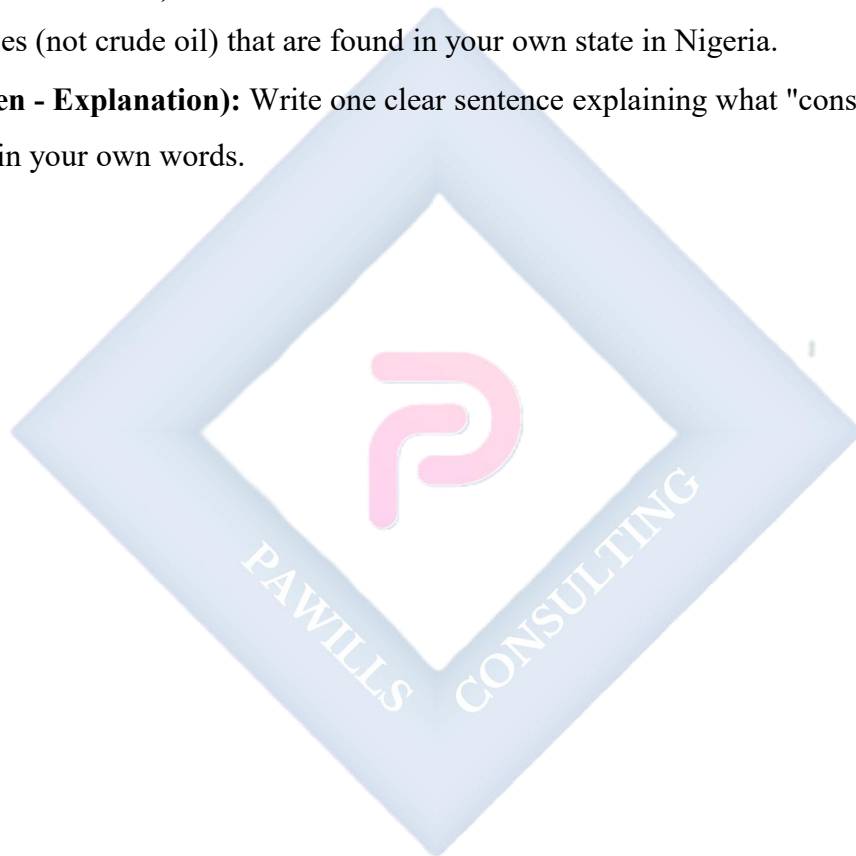
EVALUATION

1. Name the three main layers of the Earth. Which layer is the thinnest?
2. What is a natural resource? Give two examples of resources that come from plants.
3. What is the difference between the mineral "gold" and the activity of "mining"?

4. Name one important mineral found in Nigeria and what it is used for.
5. Why should we not waste resources like water or cut down all the trees in a forest?
6. What does "sustainable use" of resources mean?

ASSIGNMENT

1. **(Drawing):** Draw and color a diagram showing the Earth's layers. Label the Crust, Mantle, and Core.
2. **(Written/Research):** Ask an elder or use a textbook to find out and list two natural resources (not crude oil) that are found in your own state in Nigeria.
3. **(Written - Explanation):** Write one clear sentence explaining what "conservation" means in your own words.



WEEK 8: WATER CYCLE AND CONSERVATION

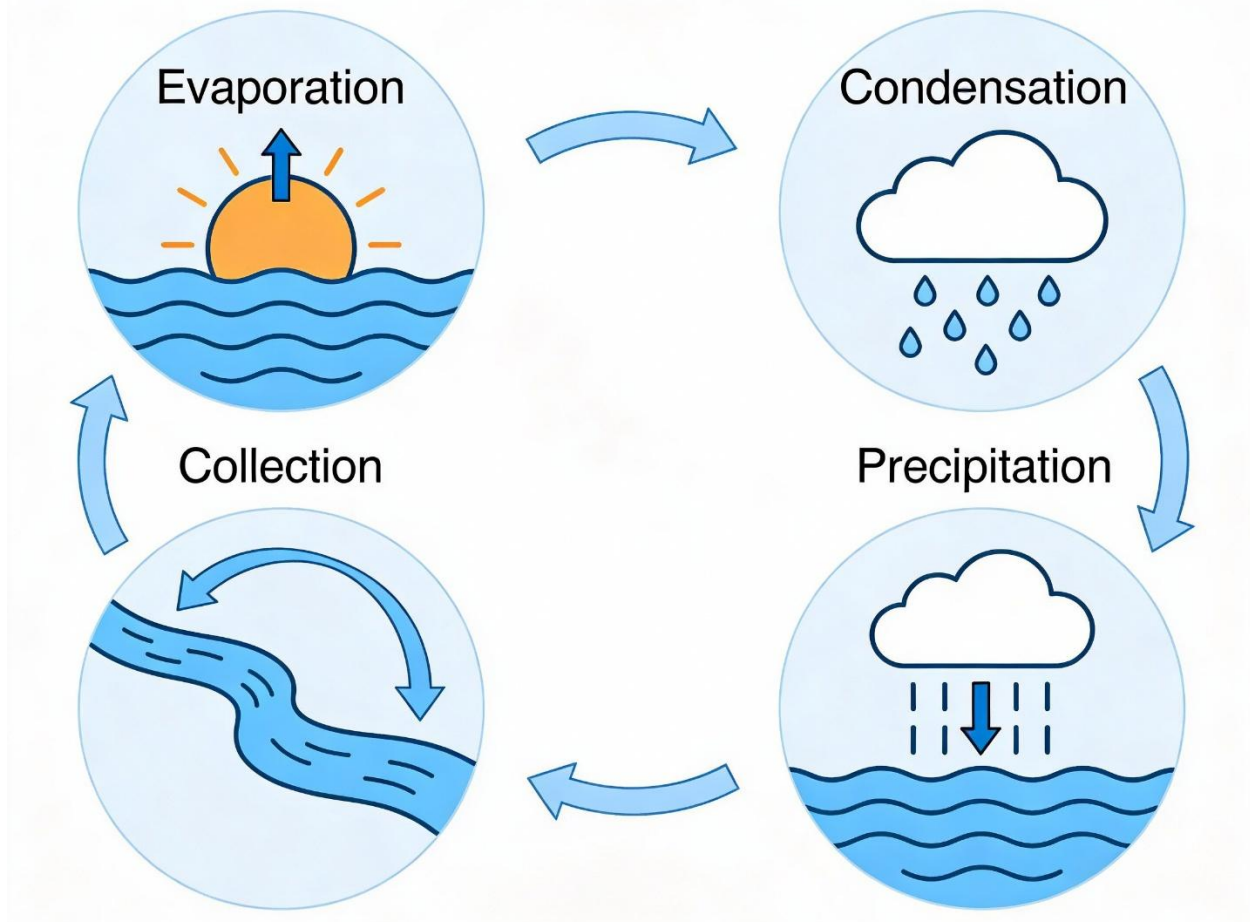
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Describe the water cycle process (Evaporation, Condensation, Precipitation).
2. State three important uses of water.
3. Explain how to treat water to make it safe.
4. Identify causes of water pollution.

CONTENT

1. The Water Cycle (Hydrological Cycle) The Earth has a fixed amount of water. The water cycle is nature's way of recycling and moving water around from the land, to the sky, and back again. It never stops.

- **Step 1: Evaporation:** The sun heats up water in rivers, lakes, oceans, and even puddles. This heat turns the liquid water into an invisible gas called water vapour, which rises up into the air.
 - You see this when a puddle dries up in the sun or when steam rises from a pot of boiling water.
- **Step 2: Condensation:** As the water vapour rises high into the cool sky, it loses heat and turns back into tiny liquid water droplets. Millions of these droplets gather together to form clouds.
 - You see this when water droplets form on the outside of a cold bottle of water or on a bathroom mirror after a hot shower.
- **Step 3: Precipitation:** When the cloud droplets get too heavy and join together, they fall back to the Earth due to gravity. This is precipitation. It can be rain, snow, hail, or sleet.
- **Step 4: Collection:** The fallen water collects on the ground in rivers, lakes, and oceans. Some soaks into the soil to become groundwater. Plants also absorb water. The cycle then starts all over again.



2. Why is Water Important? We cannot live without water.

- **For Life:** All humans, animals, and plants need water to survive. We drink it.
- **For Health and Cleanliness:** We use it for bathing, washing clothes, and cleaning our homes to stay healthy.
- **For Food:** Farmers need it to water crops (irrigation) and for animals to drink.
- **For Work and Industry:** Used in factories, for generating electricity (hydro-power), and for transportation (boats and ships).

3. Water Treatment: Making Water Safe

Not all water is safe to drink directly. It may contain germs and dirt. We can make it safe by:

- **Boiling:** Heating water until it bubbles (boils) for at least 5 minutes kills harmful germs.
- **Filtration:** Passing water through a clean cloth, ceramic filter, or sand to remove solid particles like sand and mud.

- **Chemical Treatment:** Adding a few drops of water guard (chlorine) or a small amount of alum to kill germs and make dirt settle.
- **Settling/Sedimentation:** Allowing muddy water to stand so that dirt settles at the bottom, then carefully taking the clear water from the top.

4. Water Pollution and Conservation

- **Water Pollution:** This means making water dirty, harmful, or unsafe by adding things that don't belong there.
 - **Causes:** Dumping household waste and sewage into rivers, throwing plastic bottles and nylon into water, oil spills from pipelines, washing cars and clothes directly in streams, using too much chemical fertilizer on farms (washes into rivers).
- **Water Conservation:** This means using water wisely and not wasting it.
 - **How to conserve:** Turn off taps tightly so they don't drip. Take shorter showers. Use a bucket to wash a car instead of a running hose. Collect rainwater in a barrel for watering plants. Report leaking pipes.

EVALUATION

1. What is the main source of energy that drives the water cycle?
2. During which stage of the water cycle do clouds form?
3. Give two reasons why water is important for a farmer.
4. Why is it necessary to boil or treat water from a river before drinking it?
5. What is water pollution? Give one example of how people pollute water in your community.
6. Mention one simple way you can save (conserve) water at home.

ASSIGNMENT

1. **(Drawing):** Draw your own version of the Water Cycle. Include and label: The Sun, a Cloud, Rain, and a River.

2. **(Written/Research):** Ask at home or in your community: What are two common methods people use here to make well or stream water safe for drinking? (e.g., boiling, adding alum).
3. **(Practical/Observation):** Look around your neighbourhood or on your way to school. Write down one thing you see that could lead to water pollution (e.g., a clogged drain, rubbish near a stream).



WEEK 9: AIR COMPOSITION AND QUALITY

OBJECTIVES By the end of the lesson, pupils should be able to:

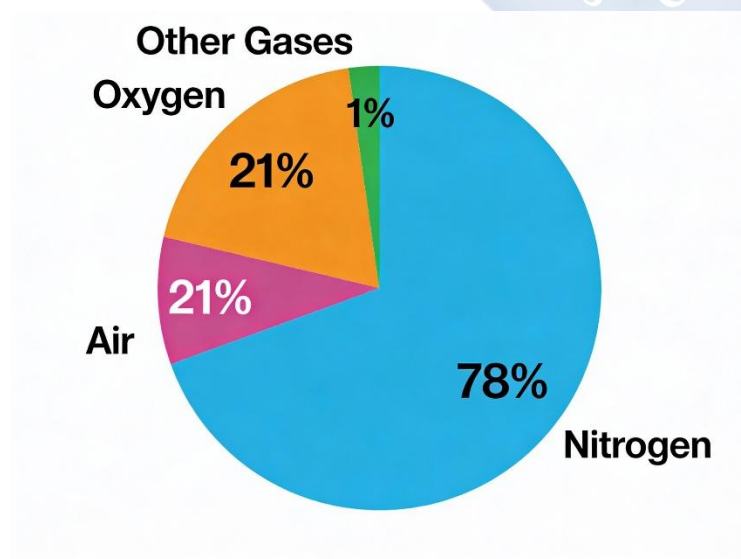
1. List the major gases found in the air.
2. State the importance of Oxygen and Carbon Dioxide.
3. Identify causes of air pollution.
4. Suggest ways to improve air quality.

CONTENT

1. What is Air? Air is a mixture of gases that surrounds the Earth. We cannot see it, but we can feel it (as wind) and we need it to breathe. It is not just "nothing"—it is matter.

2. Composition of Air Clean, dry air is made up of these main gases:

- **Nitrogen (N_2) - 78%:** This is the largest part. Most of it is not used directly by humans or animals, but it is very important for plants.
- **Oxygen (O_2) - 21%:** This is the most important gas for animals and humans. We breathe it in to survive.
- **Carbon Dioxide (CO_2) - 0.04%:** This is a very small but very important part. Plants need it to make food.
- **Other Gases - 0.96%:** Includes argon, neon, helium, and water vapour.



3. Importance of Key Gases

- **Oxygen (O₂):**
 - **For Breathing:** All animals (including humans) breathe in oxygen. Our cells use it to break down food and release energy.
 - **For Burning (Combustion):** Fire needs oxygen to burn. A candle will go out if you cover it with a jar because it uses up the oxygen inside.
- **Carbon Dioxide (CO₂):**
 - **For Photosynthesis:** Plants take in carbon dioxide from the air to make their own food (sugar).
 - **In Fire Extinguishers:** Some fire extinguishers release CO₂ to smother fire by pushing away oxygen.
- **Nitrogen (N₂):** Bacteria in the soil and in some plant roots convert nitrogen into a form plants can use to grow well. This is why fertilizers often contain nitrogen.

4. Air Pollution and Quality

- **Air Pollution:** This happens when harmful substances (pollutants) are released into the air, making it dirty and unsafe to breathe.
 - **Causes (Sources):**
 - **Smoke from vehicles:** Cars, buses, motorcycles, especially old ones with bad engines.
 - **Smoke from factories and generators:** Releases soot and poisonous gases.
 - **Burning of waste:** Burning rubbish, especially plastics, releases very toxic smoke.
 - **Dust from construction sites and unpaved roads.**
 - **Effects:** Causes coughing, asthma, eye irritation, and can lead to serious diseases. It also makes buildings dirty and affects plants.
- **Improving Air Quality (What We Can Do):**

- **Plant more trees:** Trees absorb carbon dioxide and give out oxygen. They also trap dust.
- **Reduce burning:** Avoid burning leaves or plastic waste. Dispose of trash properly.
- **Maintain vehicles:** Get cars and generators serviced regularly so they produce less smoke.
- **Use cleaner energy:** Use solar lights or improved cooking stoves that use less firewood.

EVALUATION

1. Which gas makes up most of the air we breathe?
2. Why is oxygen so important for humans and animals?
3. What important process in plants uses up carbon dioxide from the air?
4. Mention two common sources of air pollution in a town or city.
5. What percentage of the air is oxygen? (Is it more or less than nitrogen?)
6. How does planting trees help to improve air quality?

ASSIGNMENT

1. **(Drawing):** Draw a simple pie chart in your notebook to show the composition of air. Label the sections for Nitrogen and Oxygen.
2. **(Observation/Written):** On your way home from school, look around. List two things you see that can cause air pollution (e.g., a smoking vehicle, a burning heap).
3. **(Written):** Complete this sentence: "We should plant trees because they take in _____ and give out _____."

WEEK 10: WEATHER AND CLIMATE CHANGE

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Differentiate between weather and climate.
2. Identify instruments used to measure weather.
3. State the causes and effects of climate change.
4. Mention actions to fight climate change.

CONTENT

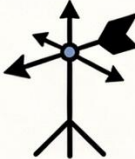
1. Weather vs. Climate

- **Weather:** This describes the condition of the atmosphere at a specific place and time. It changes from day to day, even hour to hour.
 - **Examples:** "It is sunny and hot in Lagos today." "There was a heavy rainfall in Abuja yesterday afternoon."
 - Weather tells us about temperature, rainfall, wind, cloudiness, and humidity right now.
- **Climate:** This describes the average weather conditions of a place over a very long period (30 years or more). It is the usual pattern.
 - **Examples:** "Nigeria has a tropical climate with wet and dry seasons." "The climate in the northern part of Nigeria is hot and dry (Sahel)."
 - Climate tells us what kind of weather to expect generally in a place.

2. Weather Instruments Scientists use special tools to measure and predict the weather accurately.

- **Thermometer:** Measures temperature (how hot or cold it is). It uses a liquid (like mercury or alcohol) that expands when hot and contracts when cold.
- **Rain Gauge:** Measures the amount of rainfall in millimetres (mm). It is a funnel that collects rain into a measuring cylinder.
- **Wind Vane:** Shows the direction from which the wind is blowing (North, South, East, West). The arrow points into the wind.

- **Anemometer:** Measures wind speed. It has cups that spin faster when the wind is strong.
- **Hygrometer:** Measures humidity (the amount of water vapour in the air).

Instrument	Picture	What it Measures
Thermometer		Temperature
Rain Gauge		Rainfall amount
Wind Vane		Wind direction
Anemometer		Wind speed

3. Climate Change and Global Warming

- **Climate Change:** This refers to a significant change in the Earth's average climate patterns over a long period. The Earth's climate is now changing much faster than usual.
- **Global Warming:** This is the gradual increase in the Earth's average temperature caused by an increase in certain gases in the atmosphere, called greenhouse gases (like carbon dioxide).
- **Causes of Climate Change (Why it's happening):** i. **Burning Fossil Fuels:** Burning petrol, diesel, coal, and gas for energy (in cars, generators, factories) releases huge amounts of carbon dioxide. ii. **Deforestation:** Cutting down large areas of forests (like rainforests) means there are fewer trees to absorb carbon dioxide.

- **Effects of Climate Change:**

- **Rising Temperatures:** More frequent and intense heatwaves.
- **Changing Rainfall Patterns:** Unpredictable rains, causing floods in some places and droughts (no rain) in others, which affects farming.
- **Melting Ice and Rising Sea Levels:** Ice at the poles is melting, causing oceans to rise. This can lead to flooding in coastal cities like Lagos.
- **Stronger Storms:** More powerful hurricanes and cyclones.

4. What We Can Do (Actions to Fight Climate Change) Even as children, we can help!

- **Plant Trees:** This is one of the best things to do. Join school tree-planting clubs.
- **Conserve Energy:** Turn off lights, fans, and TVs when not in use. This reduces the need to burn fuel for electricity.
- **Use Renewable Energy:** Encourage the use of solar panels and solar lamps.
- **Reduce, Reuse, Recycle:** This saves energy and resources used to make new things.
- **Spread Awareness:** Tell your family and friends about climate change.

EVALUATION

1. What is the difference between saying "The weather is rainy" and "The climate is tropical"?
2. Which instrument would a farmer use to find out how much rain fell on his farm?
3. What does a wind vane tell us about the wind?
4. What is the main cause of global warming? (Hint: Think of gases from cars and factories).
5. Name one negative effect of climate change on farming in Nigeria.
6. How does planting trees help to fight climate change?

ASSIGNMENT

1. **(Drawing):** Choose one weather instrument (e.g., a wind vane or a rain gauge) and draw a simple, labeled picture of it.

2. **(Practical/Record Keeping):** For the next three days, observe and record the weather in a small table. Columns: Day, Morning (Sunny/Cloudy/Rainy), Afternoon (Sunny/Cloudy/Rainy).
3. **(Written - Persuasive):** Write one sentence that could be on a poster, advising people in your community not to cut down trees without planting new ones.



WEEK 11: REVISION

CONTENT

- **Review of Energy:** Recap Forms of Energy (Mechanical, Heat, Light, etc.), Energy Transformation (e.g., torch, generator), and Renewable vs. Non-Renewable sources (Solar vs. Petrol).
- **Review of Electricity & Magnetism:** Recap simple circuits (closed/open), Series vs. Parallel wiring, uses of magnets, and electromagnets.
- **Review of Force & Motion:** Recap types of forces (Contact/Non-contact), Effects of Force, Friction (good and bad), and the difference between Speed and Velocity.
- **Review of Work, Energy & Power:** Recap the scientific meaning of Work (Force x Distance), its relationship to Energy, and why machines are never 100% efficient.
- **Review of Earth & Resources:** Recap Earth's layers (Crust, Mantle, Core), Natural Resources (water, minerals, trees), and the importance of conservation.
- **Review of Water & Air:** Recap the Water Cycle stages, importance and conservation of water. Recap Air Composition (Nitrogen, Oxygen, CO₂), air pollution, and air quality.
- **Review of Weather & Climate:** Recap the difference between Weather and Climate, common weather instruments, and the causes/effects of Climate Change.
- **Practice Questions for End of Term:** Create mixed questions that require applying knowledge from different topics. Example: "How does burning firewood (Energy topic) contribute to both air pollution (Air topic) and climate change (Weather topic)?"

PRIMARY 5 BASIC SCIENCE & TECHNOLOGY SCHEME OF WORK

THIRD TERM

Week	Topic	Content
1	Machines in Our Lives	Complex machines; Mechanical advantage; Efficiency; Applications of machines in industry and agriculture
2	Technology and Innovation	History of technology; Modern inventions; Computer basics; Internet and communication; Coding fundamentals
3	Biotechnology and Agriculture	Modern farming methods; Crop improvement; Pest control; Organic farming; Food security
4	Space Exploration	Planets and their characteristics; Space technology; Satellites and their uses; Famous space missions; Astronomy
5	MIDTERM EXAMINATION	Assessment of topics covered in weeks 1-4
6	MIDTERM BREAK	Rest and relaxation period
7	Health and Disease	Microorganisms - bacteria, viruses, fungi; Infectious diseases; Disease prevention; Vaccination; Antibiotics
8	Nutrition and Fitness	Detailed study of nutrients; Deficiency diseases; Balanced meal planning; Exercise and fitness; Healthy lifestyle
9	Reproduction in Living Things	Asexual and sexual reproduction; Human reproduction system; Growth and development; Puberty changes; Family life education
10	Environmental Conservation	Ecosystems and biodiversity; Endangered species; Conservation methods; National parks; Individual responsibility
11	REVISION	Review of all topics covered; Practice exercises; Question and answer sessions
12	EXAMINATION	End of term assessment
13	CLOSING	Report card distribution; Vacation assignment

PRIMARY 5 BASIC SCIENCE & TECHNOLOGY LESSON NOTES

THIRD TERM

WEEK 1: MACHINES IN OUR LIVES

OBJECTIVES By the end of the lesson, pupils should be able to:

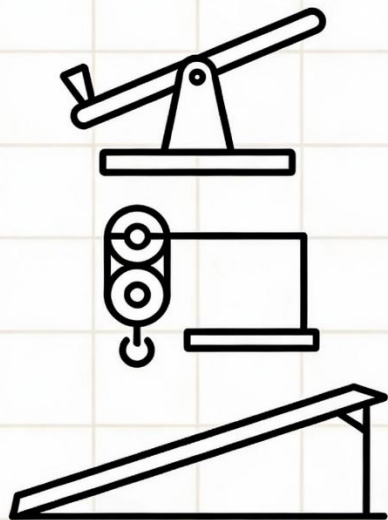
1. Define a machine and differentiate between simple and complex machines.
2. Explain the meaning of Mechanical Advantage in simple terms.
3. Define efficiency of a machine.
4. List three applications of machines in industry and agriculture.

CONTENT

1. What is a Machine? A machine is any device that makes work easier, faster, or more convenient for us. Machines help us by changing the size, direction, or speed of a force.

- **Simple Machines:** These are the basic tools that have very few or no moving parts. They are the building blocks for more complex machines. There are six main types: the lever (e.g., crowbar, see-saw), pulley (e.g., well with a bucket), inclined plane (ramp), wheel and axle (e.g., door knob, bicycle wheel), screw, and wedge (e.g., knife, axe).
- **Complex (Compound) Machines:** These are made by combining two or more simple machines. They often have many moving parts and may use electricity or fuel. They make work much easier.
 - **Examples:** A bicycle (has wheels, levers in brakes, gears). A sewing machine (has wheel, lever, screw). A car (has many simple machines working together with an engine). A grinding machine (used to grind pepper or beans).

Simple Machines



Complex Machines



2. Mechanical Advantage This sounds complicated, but it is a simple idea. Mechanical Advantage (MA) tells us how many times a machine multiplies the effort (force) we put in.

- **Simple Explanation:** If a machine has a mechanical advantage of 3, it means the machine makes your effort three times stronger. You can lift something three times heavier than you could without the machine.
- **Example 1:** Using a bottle opener (a lever) makes it very easy to remove a bottle cap. The opener gives you a mechanical advantage over trying to open it with your bare hands.
- **Example 2:** A car jack allows one person to lift a very heavy car. The jack multiplies the person's small pushing force into a huge lifting force.

3. Efficiency of a Machine No machine is perfect. Efficiency is a measure of how well a machine converts the work you put into it (input) into useful work you get out (output).

- **Why no machine is 100% efficient:** Some of the input energy is always "wasted" or lost, mostly as heat and sound due to friction between moving parts.
- **Example:** When you ride a bicycle, not all your pedaling energy goes into moving forward. Some is lost as heat in the chain and gears (you might feel them get warm) and as sound.
- **Improving Efficiency:** We can make machines more efficient by oiling and greasing moving parts to reduce friction, and by keeping them in good repair.

4. Applications of Machines Machines are everywhere and are very important for our modern life.

- **In Agriculture:**
 - Tractors for ploughing large fields.
 - Harvesters for cutting and gathering crops like rice.
 - Irrigation pumps for watering farms.
- **In Industry (Factories):**
 - Conveyor belts for moving products in a bottling plant.
 - Cranes for lifting heavy steel beams at construction sites.
 - Robotic arms for assembling cars or phones very precisely.

EVALUATION

1. What is a complex machine? Give one example.
2. A cutlass is a simple machine (wedge). A tractor is a complex machine. Explain the difference.
3. In your own words, what does "mechanical advantage" mean?
4. Why can a machine never be 100% efficient?
5. Name one machine that helps farmers plant or harvest crops.
6. What machine in a factory might be used to lift a heavy container onto a truck?

ASSIGNMENT

1. **(Drawing):** Draw a picture of a common complex machine: a bicycle. Label two parts that you think are simple machines (e.g., wheels, brake lever).
2. **(Practical/Written):** Look around your home. List three complex machines you can find (e.g., blender, ceiling fan, wristwatch).
3. **(Written - Explanation):** Why do we need to apply grease or oil to the chain of a bicycle? How does this help the bicycle?



WEEK 2: TECHNOLOGY AND INNOVATION

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define technology and mention early forms of technology.
2. Identify modern inventions that have changed our lives.
3. State the basic parts of a computer.
4. Explain the importance of the Internet and Coding basics.

CONTENT

1. What is Technology? Technology is the use of scientific knowledge and skills to create tools, machines, and systems to solve human problems and make life easier. Technology is not just electronics; it is any useful tool or method.

- **Early (Traditional) Technology:** This is how people solved problems long ago.
 - **Examples:** Using sharp stones as knives and axes, discovering how to make and control fire for cooking and warmth, inventing the wheel for transportation, making clay pots for storage.
- **Modern Technology:** This builds on old knowledge with new discoveries, especially using electricity and computers.
 - **Examples:** Electricity for light and power, Telephones for communication, Aeroplanes for fast travel, Computers for information processing.

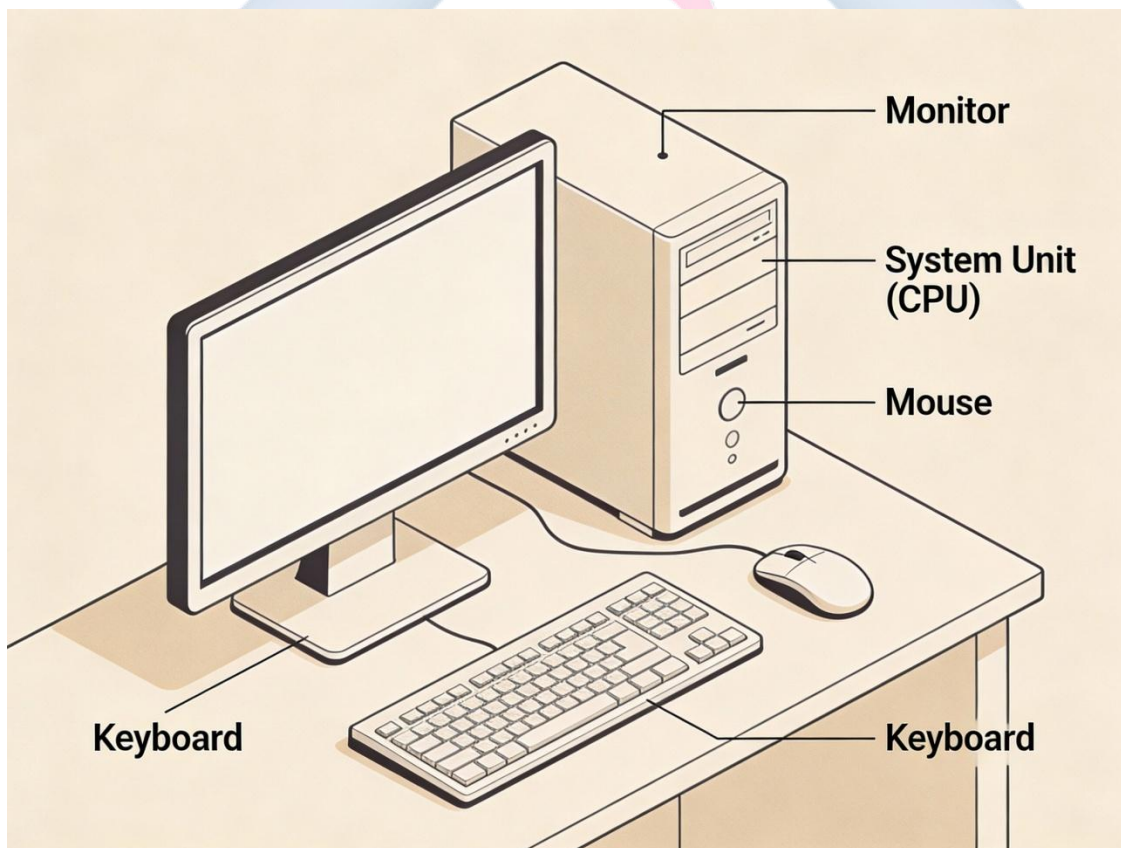
2. Modern Inventions That Changed the World

- **Communication:** Mobile Phones (Smartphones) allow us to talk, send messages, take photos, and access information from anywhere. Before phones, people sent letters which took days or weeks.
- **Transportation:** Cars, Aeroplanes, and Electric Trains make travel between cities and countries fast and convenient.
- **Health:** X-ray machines and Ultrasound scanners allow doctors to see inside the body without surgery, helping them diagnose illnesses.

- **Home Appliances:** Electric cookers, Refrigerators, and Washing machines save a lot of time and hard work at home.

3. Basic Parts of a Computer A computer is an electronic machine that accepts data (input), processes it, and produces information (output).

- **Hardware:** The physical parts you can see and touch.
 - **System Unit (CPU):** The "brain" of the computer where processing happens.
 - **Monitor:** The screen that shows information (like a TV).
 - **Keyboard:** Used for typing words and numbers (input).
 - **Mouse:** A pointing device used to click and select items on the screen.
- **Software:** The set of instructions (programs) that tell the hardware what to do. You cannot touch software.
 - **Examples:** Operating System (like Windows), Word Processors (for typing), Educational Games, Internet Browsers.



4. Internet and Coding

- **The Internet:** A worldwide network that connects millions of computers together. Think of it as a giant web of information.
 - **Uses:** Sending emails, watching videos, searching for information on Google, video calls with family far away.
- **Coding (Programming):** This is how we talk to computers. Coding means writing instructions in a special language (like Scratch or Python) that a computer can understand and follow to create games, apps, websites, and control robots.
 - **Simple Analogy:** Giving a very clear, step-by-step recipe to a robot chef. Coding is giving step-by-step commands to a computer.

EVALUATION

1. What is technology? Give one example of traditional technology.
2. How has the mobile phone changed the way we communicate compared to the past?
3. Name two parts of a computer that are used for putting information INTO the computer (input devices).
4. What is the difference between computer hardware and software?
5. Name two things you can use the internet for.
6. What is coding?

ASSIGNMENT

1. **(Drawing):** Draw and label a computer keyboard and a mouse.
2. **(Written/Interview):** Ask a parent or grandparent: "What did you use to send a message to someone far away before mobile phones were common?" Write down their answer (e.g., Letter, Telegram).
3. **(Practical/Written):** List three useful things you can do with a smartphone besides making calls.

WEEK 3: BIOTECHNOLOGY AND AGRICULTURE

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define biotechnology in simple terms.
2. Describe modern farming methods.
3. Explain the importance of crop improvement and pest control.
4. Differentiate between organic farming and chemical farming.

CONTENT

1. What is Biotechnology? Biotechnology means using living things (like tiny bacteria, yeast, or parts of plants) or their products to make or improve products for our use. It is technology based on biology.

- **Simple Example 1:** Using yeast (a tiny living fungus) to make bread rise.
- **Simple Example 2:** Using bacteria to turn milk into yoghurt or cheese.
- In farming, biotechnology helps scientists develop better seeds and plants.

2. Modern Farming Methods Farming has moved from relying only on hoes and cutlasses (subsistence farming) to using science and machines.

- **Mechanization:** Using machines like tractors, planters, and harvesters to do work faster and on larger areas of land than humans or animals could.
- **Improved Seeds (Crop Improvement):** Scientists create new varieties of seeds that are better. These seeds might:
 - Grow faster and produce more yield (more grains per plant).
 - Resist diseases that kill normal plants.
 - Survive with less water (drought-resistant), which is very important in some parts of Nigeria.
 - **Example:** NERICA Rice (New Rice for Africa) was developed to grow well in African conditions.

3. Pest and Disease Control Pests (like insects, birds, rats) and plant diseases can destroy a farmer's hard work.

- **Chemical Control:** Spraying pesticides (chemicals) to kill insects or herbicides to kill weeds. This is effective but can be harmful to other helpful insects, the soil, and human health if not used carefully.
- **Biological Control:** This is a natural method. It involves using the pest's natural enemies to control them.
 - **Example:** Releasing ladybirds to eat aphids (small insects that harm plants).
Having cats on a farm to catch rats.

4. Organic Farming vs. Chemical Farming & Food Security

- **Organic Farming:** This method grows crops without using artificial chemical fertilizers or pesticides. Instead, it uses:
 - **Natural fertilizers** like animal manure (from goats, chickens) and compost (from decayed plants).
 - **Natural methods** for pest control (like biological control).
 - Organic food is often considered healthier.
- **Chemical (Conventional) Farming:** Uses artificial fertilizers and pesticides to help crops grow fast and protect them.
- **Food Security:** This means making sure that all people, at all times, have enough safe and nutritious food to live a healthy life. Modern farming methods aim to increase food production to achieve food security for Nigeria's growing population.

EVALUATION

1. What is a simple example of biotechnology used in the kitchen? (Hint: making bread or yoghurt).
2. Why is using a tractor better than using a hoe on a very large farm?
3. Give two reasons why scientists work on improving crop seeds.
4. What is the difference between chemical and biological pest control? Which one uses natural enemies?
5. What do organic farmers use instead of artificial fertilizer?
6. What does "food security" mean?

ASSIGNMENT

1. **(Drawing):** Draw a picture of a healthy maize plant with big cobs.
2. **(Written):** List two advantages of using modern machines in farming.
3. **(Research/Oral):** Find out from a farmer or an elder at home: What is one common pest that attacks maize or bean plants in our area? (e.g., weevils, grasshoppers).



WEEK 4: SPACE EXPLORATION

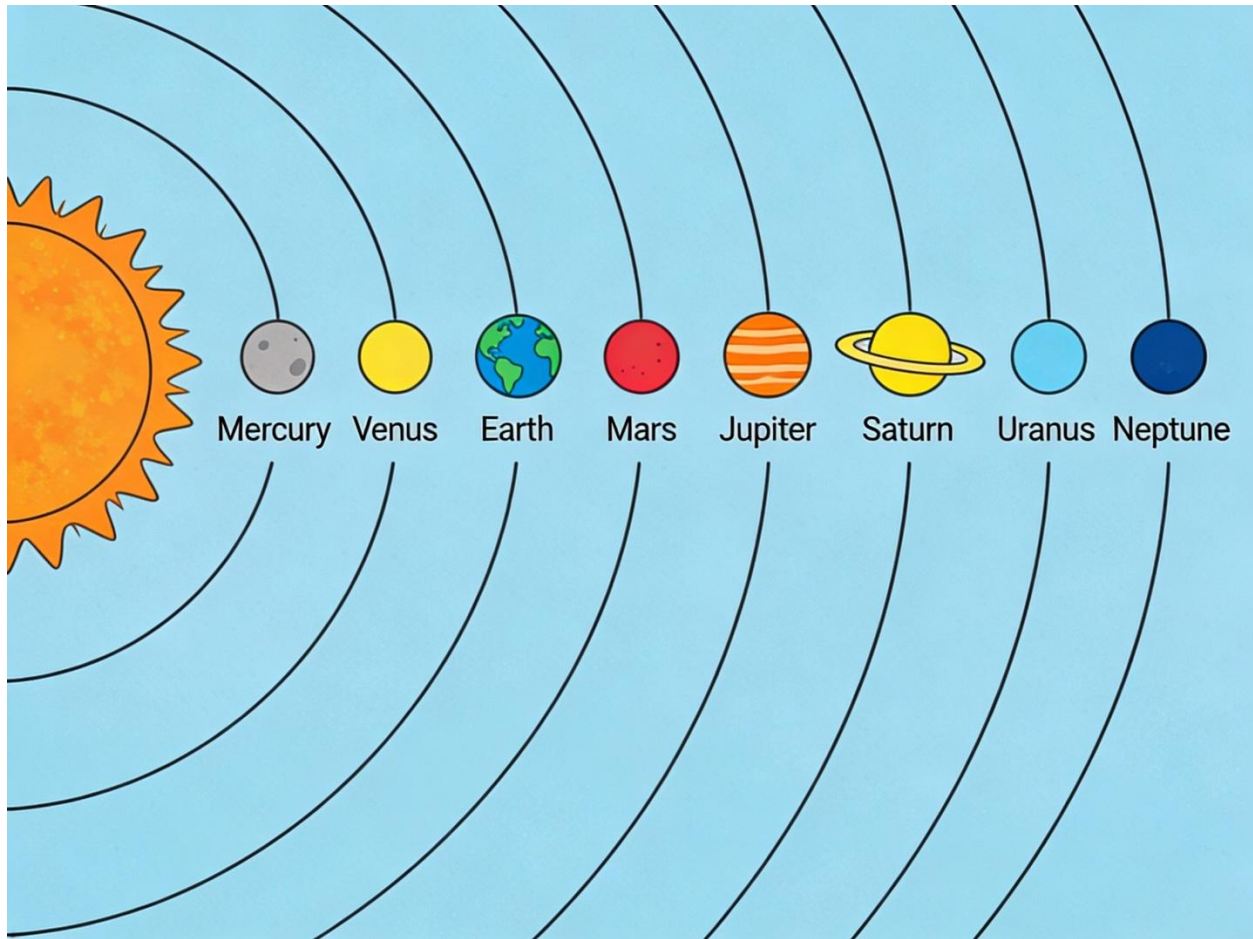
OBJECTIVES By the end of the lesson, pupils should be able to:

1. List the planets in the solar system.
2. Explain what space exploration means.
3. State the uses of satellites.
4. Mention one famous space mission or astronaut.

CONTENT

1. Our Solar System Our Solar System is made up of the Sun and all the objects that orbit (go around) it. This includes 8 planets, their moons, asteroids, and comets. The Sun is a star at the centre.

- **The 8 Planets (Order from the Sun):** A good way to remember: My Very Educated Mother Just Served Us Noodles.
 - **Mercury:** Closest to the Sun. Very hot during the day, very cold at night.
 - **Venus:** The hottest planet. It has thick, poisonous clouds.
 - **Earth:** Our home! The only planet known to have life, water, and oxygen.
 - **Mars:** The "Red Planet." Scientists send robots (rovers) here to search for signs of past life.
 - **Jupiter:** The largest planet. It has a giant red spot, which is a huge storm.
 - **Saturn:** Famous for its beautiful rings made of ice and rock.
 - **Uranus:** Orbits on its side. A very cold, gas planet.
 - **Neptune:** The windiest planet. Very cold and blue.



2. What is Space Exploration? Space Exploration is the discovery and investigation of outer space using telescopes, satellites, and spacecraft. It is about learning what is beyond our Earth.

- **Rockets:** These are powerful vehicles that can launch satellites and spacecraft into space.
- **Spacecraft and Probes:** Unmanned vehicles sent to study other planets and send back pictures and data.
- **Space Station:** A large satellite where astronauts can live and work in space for months. The International Space Station (ISS) orbits Earth.

3. Satellites and Their Uses A satellite is any object that orbits a planet. Earth has two kinds:

- **Natural Satellite:** Our Moon. It orbits the Earth.
- **Artificial Satellites:** Machines built by humans and launched into space to orbit Earth. They are very useful:

- **Communication Satellites:** Allow us to watch satellite TV (like DSTV), make international phone calls, and use the internet.
- **Weather Satellites:** Take pictures of clouds and storms from space to help forecast the weather.
- **Navigation Satellites (GPS):** Help us find locations on maps (like on our parents' phones) and guide aeroplanes and ships.
- **Earth Observation Satellites:** Monitor forests, oceans, and pollution.

4. Famous Space Missions

- **Apollo 11 Mission (1969):** This was the first mission to land humans on the Moon. Astronaut Neil Armstrong became the first person to walk on the Moon. He said the famous words: "That's one small step for man, one giant leap for mankind."
- **Astronomy:** This is the science of studying stars, planets, and other objects in space. We use telescopes to see them better.

EVALUATION

1. Name the planet we live on. What makes it special for life?
2. Which planet is the largest in our solar system?
3. What is the name of our natural satellite?
4. What do we call human-made objects that orbit the Earth?
5. How do weather satellites help us in our daily lives?
6. Who was the first person to walk on the Moon?

ASSIGNMENT

1. **(Drawing):** Draw the Solar System. Show the Sun and at least 4 planets in a line. Label Earth.
2. **(Written - Ordering):** Write down the names of the first four planets in order, starting from the one closest to the Sun.
3. **(Research):** What is the name of the tool we use on the ground to look at stars and planets? (Hint: It starts with 'T').

WEEK 5: MIDTERM EXAMINATION

(Assessment of topics covered in Weeks 1-4)



WEEK 6: MIDTERM BREAK

(Rest and relaxation period)



WEEK 7: HEALTH AND DISEASE

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define microorganisms and list three types.
2. Explain what infectious diseases are.
3. List ways to prevent diseases.
4. State the importance of vaccination and antibiotics.

CONTENT

1. Microorganisms (Microbes or Germs) These are tiny living things that are too small to be seen with just our eyes. We need a microscope to see them. Not all microorganisms are bad; some are very useful. But some cause diseases.

- **Bacteria:** Single-celled organisms. Some are helpful (used to make yoghurt, help digest food). Others are harmful and cause diseases like Cholera, Typhoid, and Tuberculosis (TB).
- **Viruses:** Even smaller than bacteria. They are not truly "alive" on their own; they need to enter a living cell to multiply. They cause many diseases like Common Cold, Measles, Chickenpox, and COVID-19.
- **Fungi:** Some are large (like mushrooms), but some are tiny and cause skin diseases. Examples: Ringworm, Athlete's Foot.

2. Infectious Diseases These are diseases that can spread from one person to another (they are communicable). They are caused by harmful microorganisms.

- **How they spread:**
 - **Through the Air:** When an infected person coughs or sneezes, germs can travel in tiny droplets to another person (e.g., Flu, Tuberculosis).
 - **Through Contaminated Food/Water:** Eating food or drinking water that has germs (e.g., Cholera, Typhoid).
 - **Through Insect Bites:** Mosquitoes can carry and spread the Malaria parasite when they bite.

- **Through Direct Contact:** Touching an infected person or their things (e.g., Ringworm, Chickenpox).

3. Disease Prevention (How to Stay Healthy) We can protect ourselves and others by practicing good hygiene.

- Wash your hands regularly with soap and water, especially before eating and after using the toilet.
- Drink clean, safe water. Boil or filter water if unsure.
- Sleep under an Insecticide-Treated Net (ITN) to prevent mosquito bites and malaria.
- Keep your environment clean to stop mosquitoes and flies from breeding.
- Cover your mouth and nose with a tissue or your elbow when coughing or sneezing.

4. Vaccination and Antibiotics

- **Vaccination:** This is a way to prevent a disease before you get it. A vaccine is a special medicine (usually an injection or drops) that contains a weakened or dead germ. It trains your body's defence system (immune system) to recognize and fight the real germ if it ever enters your body.
 - **Examples:** Polio vaccine, Measles vaccine, BCG vaccine (against TB). The immunization card keeps a record of all your vaccines.
- **Antibiotics:** These are special medicines used to treat infections caused by bacteria after you get sick (e.g., for Typhoid). They do not work against diseases caused by viruses, like the common cold or flu.

EVALUATION

1. What are microorganisms? What tool do we use to see them?
2. Name the three main types of microorganisms.
3. Which type of microorganism causes Malaria? (Is it spread by mosquito?)
4. What are two ways to prevent the spread of diarrhoea diseases like cholera?
5. What is the purpose of a vaccine? Does it treat or prevent a disease?
6. If you have a common cold caused by a virus, will antibiotics cure it? Why or why not?

ASSIGNMENT

1. **(Drawing):** Draw a bar of soap and a picture of running water. Write the sentence:
"Wash hands to keep germs away."
2. **(Written):** List three common diseases you know about in your community.
3. **(Practical):** Ask your parents to see your immunization card. Write down the name of one vaccine that is recorded on it.



WEEK 8: NUTRITION AND FITNESS

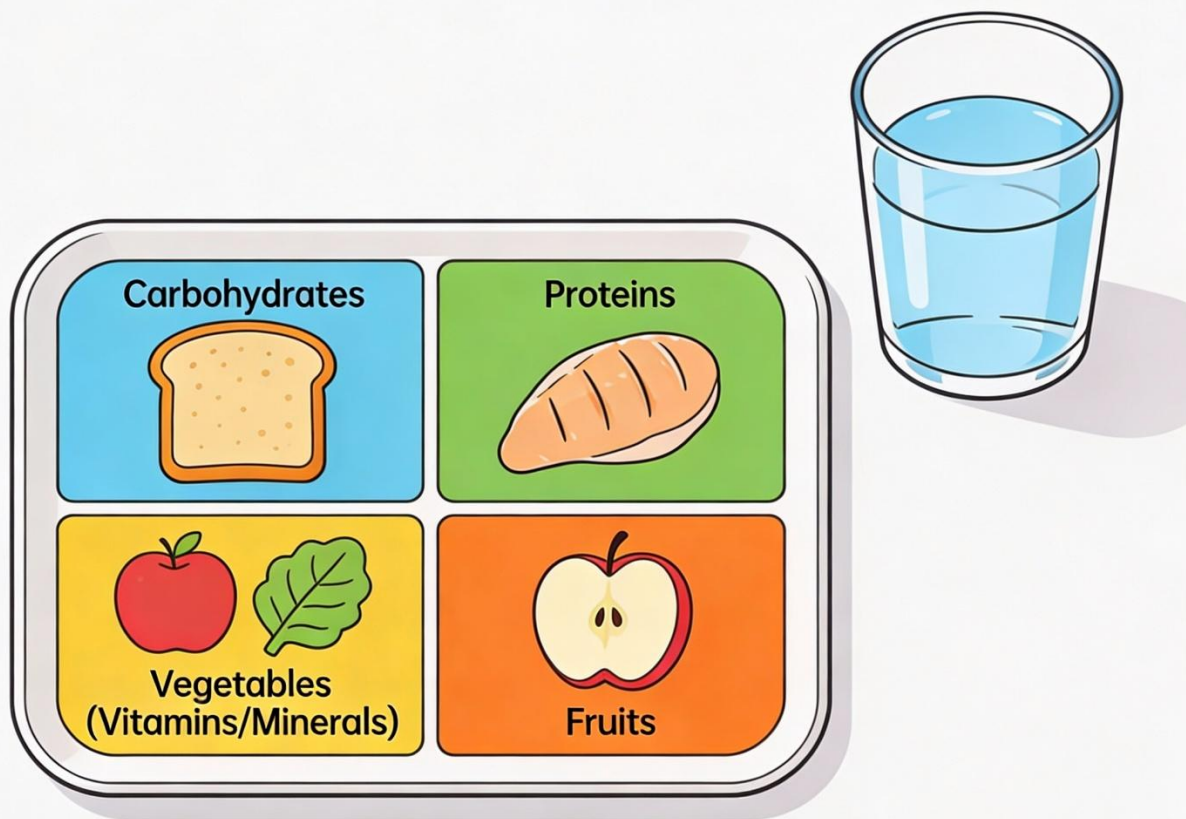
OBJECTIVES By the end of the lesson, pupils should be able to:

1. List the six classes of food nutrients.
2. Identify deficiency diseases caused by lack of specific nutrients.
3. Plan a simple balanced meal.
4. Explain the benefits of exercise and fitness.

CONTENT

1. The Six Classes of Food Our bodies need different types of nutrients from food to grow, stay healthy, and have energy. These nutrients are grouped into six classes.

- **Carbohydrates (Energy-giving foods):** They are the body's main source of fuel for energy. Examples: Garri, Rice, Yam, Bread, Maize, Potatoes.
- **Proteins (Body-building foods):** They are used for growth, to repair worn-out tissues, and to build muscles. Examples: Beans, Fish, Meat, Egg, Milk, Soybeans.
- **Fats and Oils (Also energy-giving, but richer):** They provide a lot of energy, keep the body warm, and protect organs. Examples: Palm oil, Groundnut oil, Butter, Avocado.
- **Vitamins (Protective foods):** They keep the body working properly and help fight diseases. Examples: Vitamin A (in carrots, leafy greens) for good eyesight. Vitamin C (in oranges, mangoes, guava) for healthy skin and gums.
- **Minerals (Protective foods):** They are needed for strong bones, teeth, and blood. Examples: Calcium (in milk, ugu) for bones. Iron (in liver, spinach) for blood.
- **Water:** It is not a food but is the most important nutrient. It helps digest food, carries nutrients in the blood, and cools the body through sweat.



2. Deficiency Diseases These are illnesses that occur when the body does not get enough of a particular nutrient over a long time.

- **Kwashiorkor:** Caused by severe lack of protein. Signs: Swollen stomach, thin arms and legs, discoloured hair.
- **Rickets:** Caused by lack of Vitamin D and Calcium. Signs: Weak, soft bones that bend (bow legs).
- **Scurvy:** Caused by lack of Vitamin C. Signs: Bleeding gums, slow healing of wounds.
- **Night Blindness:** Caused by lack of Vitamin A. Signs: Difficulty seeing in dim light or at night.
- **Anaemia:** Caused by lack of Iron. Signs: Weakness, tiredness, pale skin.

3. Balanced Diet and Meal Planning A balanced diet is a meal that contains the right amounts of all six classes of food in the correct proportions. Eating only one type of food (like only rice) is not balanced.

- **Example of a simple balanced meal:** Rice and Stew (Carb + Vitamins) + Fish (Protein) + Ugu Soup (Vitamins & Minerals) + A glass of water + An orange for dessert (Vitamin C).

4. Exercise and Fitness Good health is not just about food; it's also about keeping the body active. Exercise is any activity that makes your body work harder than normal.

- **Benefits of Regular Exercise:**
 - Makes the heart and lungs stronger.
 - Builds strong muscles and bones.
 - Helps to maintain a healthy weight.
 - Makes you feel happy and energetic.
 - Helps you sleep better.
- **Examples of Exercise:** Running, Jumping rope (skipping), Playing football, Dancing, Swimming, Cycling.

EVALUATION

1. Which class of food is our main source of energy? Give one example.
2. Why do growing children need plenty of protein-rich foods?
3. What deficiency disease is linked to a lack of Vitamin C?
4. What are the two main causes of Kwashiorkor? (Think of the nutrient and the common situation).
5. Why is drinking enough water important for the body?
6. Mention two benefits of regular exercise.

ASSIGNMENT

1. **(Written - Planning):** Create a menu for one balanced meal you would like to eat. List the food items and state the main nutrient each one provides (e.g., Rice - Carbohydrate).

2. **(Written):** List three fruits that are good sources of Vitamin C.
3. **(Drawing):** Draw a picture of a child doing a healthy exercise you enjoy (e.g., skipping, playing football).



WEEK 9: REPRODUCTION IN LIVING THINGS

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Differentiate between asexual and sexual reproduction.
2. Identify the male and female reproductive cells in humans.
3. List changes that occur during puberty.
4. Discuss the basics of family life education.

CONTENT

1. Asexual and Sexual Reproduction Reproduction is the process by which living things produce new individuals of their own kind (babies, young ones, or new plants).

- **Asexual Reproduction:** Only one parent is needed. The offspring is identical to the parent. Common in plants and simple organisms.
 - **Example in Plants:** Growing a new plant from a stem cutting (e.g., cassava, hibiscus). Growing yam from a yam tuber.
- **Sexual Reproduction:** Involves two parents – a male and a female. Their special reproductive cells join to form the offspring, which is a mix of both parents.
 - Common in most animals and flowering plants.
 - **Example:** A hen (female) and a rooster (male) produce chicks. A woman and a man have a baby.

2. Human Reproduction (Basic Introduction) In humans, sexual reproduction involves special cells:

- **Male Reproductive Cell:** Sperm (produced by the father).
- **Female Reproductive Cell:** Egg (Ovum) (produced by the mother).
- **Fertilization:** When a sperm cell joins with an egg cell, fertilization happens. This fertilized egg then grows into a baby inside the mother's womb over nine months.

3. Puberty: The Age of Change Puberty is the time when a child's body begins to develop and change into the body of an adult. These changes are normal and natural, caused by hormones.

- **Changes in Boys:**

- Voice becomes deeper.
- Shoulders broaden.
- Hair grows on the face (beard), underarms, and pubic area.
- Production of sperm begins.

- **Changes in Girls:**

- Breasts develop.
- Hips become wider.
- Hair grows underarms and in the pubic area.
- Menstruation (monthly period) begins, which means the body is now able to have a baby.

- **Changes in Both:** Growth spurts (become taller quickly), skin may become oilier (leading to pimples), and emotions may change.

4. Family Life Education (Key Messages)

- **Hygiene is Very Important:** Bathing regularly, using deodorant, and keeping private parts clean is essential during puberty.
- **Respect Your Body and Others:** The changes are private. It is important to respect your own body and the bodies of others.
- **It's a Normal Process:** Everyone goes through puberty at their own pace. It's nothing to be afraid or ashamed of.
- **Talk to Trusted Adults:** If you are confused or worried about the changes, talk to a parent, teacher, or school nurse.

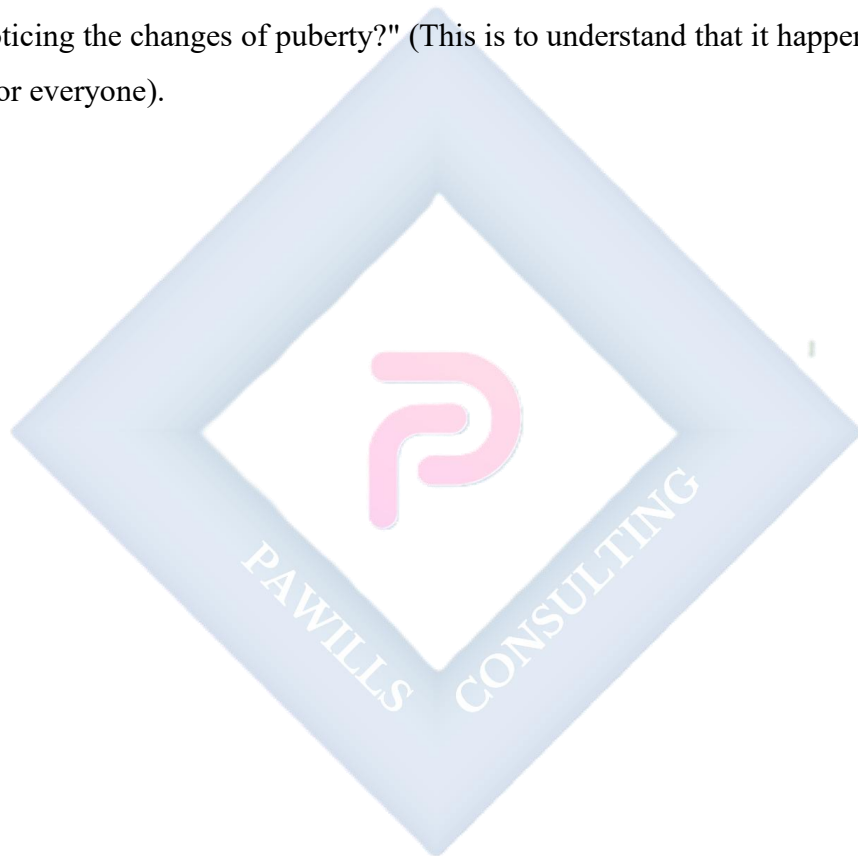
EVALUATION

1. What is the main difference between asexual and sexual reproduction?
2. Give one example of asexual reproduction in plants.
3. What is the name of the female reproductive cell?
4. Mention two body changes that happen to a girl during puberty.
5. Mention two body changes that happen to a boy during puberty.

6. Why is personal cleanliness especially important during puberty?

ASSIGNMENT

1. **(Written):** List three personal hygiene items that a teenager should use (e.g., soap, deodorant, sanitary pads for girls).
2. **(Written - Reflection):** Write one sentence about why it is important to respect our own bodies and the bodies of our classmates.
3. **(Oral/Interview):** You can ask a trusted parent or older sibling: "At what age did you start noticing the changes of puberty?" (This is to understand that it happens at different times for everyone).



WEEK 10: ENVIRONMENTAL CONSERVATION

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define ecosystem and biodiversity.
2. Explain what "endangered species" means.
3. List methods of conserving nature.
4. State the importance of National Parks.

CONTENT

1. Ecosystems and Biodiversity

- **Ecosystem:** A community where living things (plants, animals, microorganisms) interact with each other and with the non-living things (sunlight, air, water, soil) in their environment.
 - **Examples:** A pond ecosystem, a forest ecosystem, a grassland ecosystem.
- **Biodiversity:** This word comes from "biological diversity." It means the variety of different types of life found in an ecosystem or on Earth. It includes all the different plants, animals, fungi, and even tiny microbes.
 - **Why it's good:** High biodiversity makes an ecosystem strong and healthy. It's like a football team needing players in different positions.

2. Endangered Species An endangered species is a type of plant or animal that is at serious risk of disappearing forever (becoming extinct). This is usually because their population has become very small.

- **Causes:** Habitat loss (forests being cut down), poaching (illegal killing for body parts like ivory or skin), pollution, and climate change.
- **Examples of Endangered Animals in Africa:** Elephants (killed for ivory tusks), Gorillas, Rhinoceros, Pangolins, Sea Turtles.



3. Methods of Conservation Conservation means protecting and managing our environment and the living things in it so they can continue to exist. We can conserve by:

- **Afforestation and Reforestation:** Planting new trees to replace those cut down.
- **Laws and Protected Areas:** Governments make laws against poaching and create National Parks and Game Reserves where animals are protected.
- **Recycling:** Reusing materials like plastic, paper, glass, and metal to reduce waste and the need to cut down more trees or mine more materials.
- **Individual Actions:** Not littering, saving water, using less plastic, and teaching others.

4. National Parks in Nigeria These are large areas of land set aside by the government where wild animals and plants are protected. People can visit to see wildlife, but hunting is strictly forbidden.

- **Importance:** They protect biodiversity, provide a home for endangered species, are a place for scientific study, and can attract tourists.
- **Examples in Nigeria:** Yankari Game Reserve (Bauchi State), Cross River National Park (Cross River State), Gashaka Gumti National Park (Taraba/Adamawa), Okomu National Park (Edo State).

EVALUATION

1. What does "biodiversity" mean in simple terms?
2. What does it mean if an animal is described as "endangered"?
3. Give one reason why elephants are becoming endangered.
4. What is "poaching"?
5. Can you name one National Park or Game Reserve in Nigeria?
6. As a pupil, what is one thing you can do at school to help the environment?

ASSIGNMENT

1. **(Drawing/Poster):** Design a simple poster with the message: "Do Not Kill Wild Animals. Protect Our Wildlife." Use colours and a drawing.
2. **(Practical/Written):** Look at items in your rubbish bin at home. List two things that could be recycled instead of thrown away (e.g., plastic bottle, old newspaper).
3. **(Research):** Find out and write down which Nigerian state is home to the Yankari Game Reserve.

WEEK 11: REVISION

CONTENT

- **Review of Machines:** Recap Simple vs. Complex machines, Mechanical Advantage, and Efficiency.
- **Review of Technology:** Recap Early vs. Modern technology, Computer parts (Hardware/Software), and the uses of the Internet.
- **Review of Biotechnology & Agriculture:** Recap modern farming, crop improvement, pest control, and organic farming.
- **Review of Space:** Recap the Solar System (8 planets), artificial satellites and their uses, and the Apollo 11 mission.
- **Review of Health:** Recap Microorganisms (Bacteria, Viruses, Fungi), disease prevention, and the difference between Vaccines and Antibiotics.
- **Review of Nutrition:** Recap the 6 Classes of Food, Deficiency Diseases, and the importance of a Balanced Diet and Exercise.
- **Review of Reproduction & Puberty:** Recap Asexual vs. Sexual reproduction, changes during puberty, and the importance of hygiene and respect.
- **Review of Environmental Conservation:** Recap Biodiversity, Endangered Species, Conservation methods (e.g., recycling, national parks).
- **Practice Questions for End of Session Exams:** Include application-based questions that link topics. Example: "How can biotechnology (Week 3) help achieve food security (also Week 3)?" or "Why is vaccination (Week 7) an important part of public health?"
- **Checking of Notes:** Ensure all pupils have complete and well-organized notes for all three terms.